The Wave Kinetic Equation

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We now derive the kinetic equation for the wave in presence of scattering by density and magnetic field fluctuations. From simple considerations about the incident wave and the fluctuations ordering we propose an appropriate parameter for the perturbative expansion of the Maxwell-Vlasov system of equations and we show that at the first and second order we obtain the Maxwell equations for the first and second order fields in closed form. By means of this perturbative analysis, we show the bilinear coupling between the incident field and the fluctuations field, that are treated as two independent linear waves, and we demonstrate that scattering in first Born approximation is a second order process due to this bilinear interaction. In the low temperature limit, the wave-wave interaction can be studied with an alternative fluid approach, based on the perturbative analysis of the MHD equations, that can be derived by a moments analysis of the Vlasov equations. We initially study the effects of the fluctuations on LH waves in the cold plasma limit, where all the temperature effects are neglected, so the fluid approach represents a good approximation. Also from the analysis of the MHD equations for a cold plasma it is easy to show that scattering is a second order effect. We derive an explicit expression for the second order current due to interaction between incident field and fluctuations in the cold plasma limit, starting from second order fluid equations. This expression is then inserted in the second order non homogeneous Maxwell equations, and the mode coupling equation that describes the coupling between the incident field and the fluctuations is derived in details. From this equation we can then derive the wave kinetic equation applying the random phase approximation, the wave kinetic equation describes the evolution of the wave energy density due to the scattering events in the plasma and its solution give us the evaluation of the scattering probability along the geometrical optics ray .

1 Fluctuations ordering

We now analyze the temporal and spatial scaling of the fluctuations comparing the fluctuations frequency $\widetilde{\omega}$ and the fluctuations wave vector $\widetilde{k}$ with the frequency $\omega$and the wave vector k of the incident lower hybrid wave calculated using the parameters of several tokamak devices actually having an LH current drive system and also the ITER parameters. The parameters of the tokamaks

considered in the analysis are reported in the table:

	JET	TORE SUPRA	ALCATOR	FTU	ITER
a	$2.96m$	$2.46m$	$0.67m$	$0.935m$	$6.2m$
R_0	$1.24m$	$0.68m$	$0.21m$	$0.31m$	$2m$
$B^{(0)}$	$3.45T$	$4T$	$9T$	$8T$	$5,3T$
$n_e(0)$	$3 \cdot 10^{19} m^{-3}$	$3.6 \cdot 10^{19} m^{-3}$	$3.5 \cdot 10^{20} m^{-3}$	$10^{20} m^{-3}$	$10^{20} m^{-3}$
$q(a)$	4	5	7	7	7
$f = \omega/2\pi$	$3.7GHz$	$3.7GHz$	$4.6GHz$	$8GHz$	$5GHz$

a and R_0 are respectively the minor and the major radius of the tokamaks, $B^{(0)}$is the toroidal on axis magnetic field, $n_e(0)$ is the central electron density, $q(a)$ is the safety factor at the border of the plasma, and f is the lower hybrid wave frequency.

1.1 Time ordering

As we can observe from the values reported in the table, the frequency $f = \omega/2\pi$ of the lower hybrid incident wave in tokamaks JET, TORE SUPRA and ALCATOR C-MODE is of the order of $1GHz = 10^9 Hz$, the LH system projected for ITER will works with $f = 5GHz$, so of the order of $10GHz = 10^{10} Hz$, a system that use this range of frequencies is just in operation on the FTU tokamak, in order to simulate the conditions of an LHCD discharge on ITER. In order to study the fluctuations effects in actual tokamaks scenarios, we assume f of the order of $10^9 Hz$. The fluctuations frequency $\widetilde{f} = \widetilde{\omega}/2\pi$ is $\widetilde{f} \approx 100KHz = 10^5 Hz$, [2],[4], so $\widetilde{\omega} \ll \omega$ and the fluctuations time-scale is slower than incident wave time-scale, for the time-scales we have the small ordering parameter:

$$\delta^2 = \frac{\widetilde{\omega}}{\omega} \approx \frac{10^5}{10^9} = 10^{-4} \tag{1}$$

It is important to remember that this ordering parameter is obtained considering the characteristics of actual tokamaks scenarios, if we consider the parameters of an ITER-like scenario we obtain $\delta^2 = \widetilde{\omega}/\omega \approx 10^{-5}$. We can then separate the temporal dynamics of the incident wave from the temporal dynamics of the fluctuations, the time derivative operator becomes (see appendix A):

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial t_\omega} + \delta^2 \frac{\partial}{\partial t_{\widetilde{\omega}}} \tag{2}$$

1.2 Space ordering

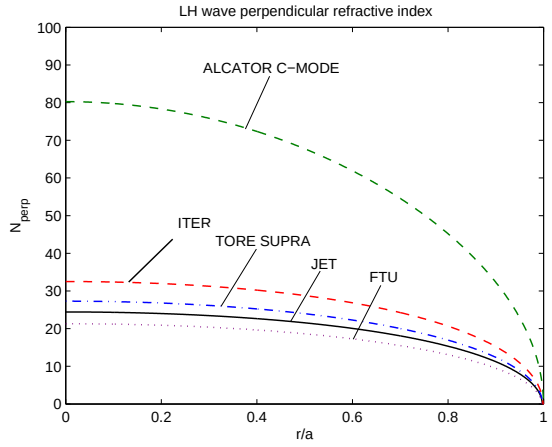
The perpendicular refractive index of the LH incident wave in the cold-electrostatic approximation is given by the dispersion relation:

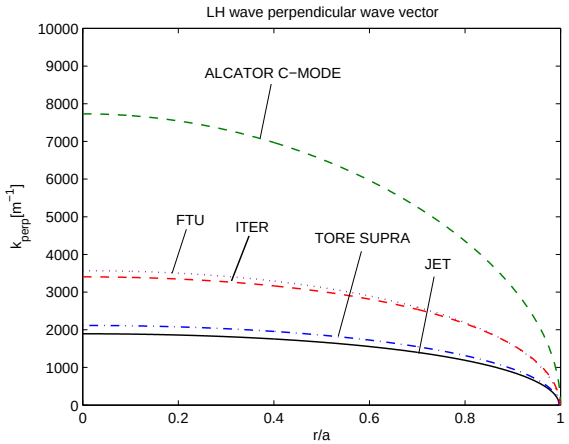
$$N_\perp^2 = -\frac{K_\parallel^C}{K_\perp^C} N_\parallel^2 \tag{3}$$

Where for a two species plasma the elements $K_\perp^C$ and $K_\parallel^C$ of the cold plasma permittivity tensor, remembering that $\omega_{ci}^2 \ll \omega^2 \ll \omega_{ce}^2$ are:

$$K_\perp^C \simeq 1 + \frac{\omega_{pe}^2}{\omega_{ce}^2} - \frac{\omega_{pi}^2}{\omega^2}$$
$$K_\parallel^C \simeq 1 - \frac{\omega_{pe}^2}{\omega^2}$$

In our considerations, we assume the costant value $N_\parallel = 2$ that is the tipical initial value of the LH waves used in actual LHCD experiments ($N_\parallel$ in general remains of order 1 into the plasma). The values of $N_\perp$ and of the perpendicular component of the wave vector $k_\perp = \omega N_\perp / c$, in function of the normalized radial coordinate, calculated with the expression (3) for the tokamak parameters reported in the table, are illustrated in the following figures:





We have assumed electron density at the border $n_e(a) = 0$, $B = cost = B^{(0)}$ and $q = cost = q(a)$. From these plots we can observe that except for the case

of ALCATOR C-MODE, that presents very high density and toroidal magnetic field, in tokamaks actually in operation as TORE SUPRA and JET and also in ITER-like scenarios, the perpendicular component of the LH wave vector is of the order of $k_\perp \approx 10^3 m^{-1}$. The values of $k_\parallel = \omega N_\parallel / c$ and of $\lambda_\parallel = 2\pi / k_\parallel$ calculated for $N_\parallel = 2$ using the tokamak parameters reported in the table, are the following:

	JET	TORE SUPRA	ALCATOR	FTU	ITER
$k_\parallel$	$155 m^{-1}$	$155 m^{-1}$	$193 m^{-1}$	$335 m^{-1}$	$209 m^{-1}$
$\lambda_\parallel$	$4.1 \cdot 10^{-2} m$	$4.1 \cdot 10^{-2} m$	$3.2 \cdot 10^{-2} m$	$1.9 \cdot 10^{-2} m$	$3 \cdot 10^{-2} m$

The modulus of the fluctuations wave vector, $\widetilde{k}$, has been measured experimentally with several diagnostic techniques[4, 5, 7]; $\widetilde{k}$ varies from values of the order of $\widetilde{k} \approx 0.005 cm^{-1} = 0.5 m^{-1}$, to values of the order of $\widetilde{k} \approx 10^3 m^{-1}$, that is of the same order of the perpendicular incident lower hybrid wave vector, $k_\perp \approx 10^3 m^{-1}$. From these experimental studies it has been observed that $\widetilde{k}_\parallel \ll \widetilde{k}_\perp$ and $\widetilde{\lambda}_\parallel \gg \widetilde{\lambda}_\perp$, so we can approximate $\widetilde{k} \simeq \widetilde{k}_\perp$ and the component of the scattered wave vector becomes $\check{k}_\parallel = k_\parallel$ and $\check{k}_\perp = k_\perp + \widetilde{k}_\perp$; for small values of $\widetilde{k}_\perp$, $\check{k} = k$ and we have like-mode scattering, for values of $\widetilde{k}_\perp$ of the same order of $k_\perp$, $\check{k} \neq k$ and we have unlike-mode scattering. From the experiments it has been demonstrated the relation $\widetilde{\lambda}_\parallel \simeq (0.1 - 0.3) q R_0$, [6, 8], and as a consequence $\widetilde{k}_\parallel \simeq 2\pi / (0.1 - 0.3) q R_0$, in the following table are reported the minimum and the maximum values of $\widetilde{k}_\parallel$and $\widetilde{\lambda}_\parallel$ calculated with these relations using the tokamak parameter of the first table:

	JET	TORE SUPRA	ALCATOR	FTU	ITER
$\widetilde{k}_{\parallel min}$	$1.8 m^{-1}$	$1.7 m^{-1}$	$4.5 m^{-1}$	$3.2 m^{-1}$	$0.48 m^{-1}$
$\widetilde{k}_{\parallel max}$	$5.3 m^{-1}$	$5.1 m^{-1}$	$11.4 m^{-1}$	$9.6 m^{-1}$	$1.44 m^{-1}$
$\widetilde{\lambda}_{\parallel min}$	$1.2 m$	$1.2 m$	$0.47 m$	$0.65 m$	$4.3 m$
$\widetilde{\lambda}_{\parallel max}$	$3.6 m$	$3.7 m$	$1.4 m$	$2.0 m$	$13 m$

From these values we can see that we have $\widetilde{k}_\parallel \ll k_\parallel$ and $\widetilde{\lambda}_\parallel \gg \lambda_\parallel$ for all set of tokamak parameters, as a consequence, we can define the small parameter $\widetilde{k}_\parallel / k_\parallel = \lambda_\parallel / \widetilde{\lambda}_\parallel$ and we can calculate the range of values of this parameter for all the tokamaks considered:

	JET	TORE SUPRA	ALCATOR	FTU	ITER
$\widetilde{k}_{\parallel min} / k_\parallel$	$1.1 \cdot 10^{-2}$	$1.1 \cdot 10^{-2}$	$2.3 \cdot 10^{-2}$	$0.9 \cdot 10^{-2}$	$2.3 \cdot 10^{-3}$
$\widetilde{k}_{\parallel max} / k_\parallel$	$3.4 \cdot 10^{-2}$	$3.3 \cdot 10^{-2}$	$6 \cdot 10^{-2}$	$2.9 \cdot 10^{-2}$	$6.8 \cdot 10^{-3}$

For parameters of tokamaks JET, TORE SUPRA and FTU, the range of possible values of $\widetilde{k}_\parallel / k_\parallel$ is of the order of 10^{-2}, for ALCATOR C-MODE parameters it is between $10^{-2}$and 10^{-1} and for ITER parameters it varies from

$10^{-3}$to 10^{-2}. So for studying the fluctuations effects in tokamaks like JET or TORE SUPRA and also in several ITER-like scenarios we can choose the ordering parameter:

$$\frac{\widetilde{k}_{\parallel}}{k_{\parallel}} = \frac{\lambda_{\parallel}}{\widetilde{\lambda}_{\parallel}} \approx 10^{-2} = \left(\frac{\widetilde{\omega}}{\omega}\right)^{\frac{1}{2}} = \delta \tag{4}$$

From this discussion it is important to remember that in tokamaks with small radius R_0 and high magnetic field $B^{(0)}$, this space-ordering parameter can reach values of the order of 10^{-1} and that in tokamaks with very big radius (ITER, DEMO) it can be of the order of 10^{-3}.

Now, using this ordering parameter, we can separate the spatial dynamics of the wave from the parallel spatial dynamics of the fluctuations, the gradient operator then becomes (see appendix B):

$$\nabla = \nabla_{\lambda} + \delta \nabla_{\parallel \widetilde{\lambda}_{\parallel}} \tag{5}$$

The symbol $\nabla_{\parallel}$means that the first order terms of the gradient operator, due to the fluctuations, is only regarding the dynamics parallel to the magnetic field.

1.3 Perturbative analysis of the equations

If we neglect the ion dynamics, $(m_i \gg m_e, \mathbf{p}_i \ll \mathbf{p}_e)$, the electron distribution function, $f_e(\mathbf{r}, \mathbf{p}, t)$ in a collisionless plasma is solution of the Vlasov kinetic equation:

$$\frac{\partial f_e}{\partial t} + \mathbf{v} \cdot \nabla f_e + q_e (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f_e}{\partial \mathbf{p}} = 0 \tag{6}$$

Where the macroscopic fields $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{B}(\mathbf{r}, t)$ are solutions of the Maxwell equations:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{7}$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \tag{8}$$

And the current is given by:

$$\mathbf{J} = q_e \int \mathbf{v} f_e \mathrm{d}\mathbf{p} \tag{9}$$

This system of equation has a set of equilibrium solutions, $f_e^{(0)}, \mathbf{E}^{(0)}, \mathbf{J}^{(0)}$and $\mathbf{B}^{(0)}$, we assume $\mathbf{E}^{(0)} = 0$ at the equilibrium, and we consider other sets of solutions of the Maxwell-Vlasov system, due to several physical phenomena, as small perturbations to the equilibrium solutions, using the ordering parameter (4) for the expansion:

$$
\begin{aligned}
f_e(\mathbf{r},\mathbf{p},t) &= f_e^{(0)} + \delta f_e^{(1)} + \delta^2 f_e^{(2)} + \dots && (10)\\
\mathbf{E}(\mathbf{r},t) &= \delta\mathbf{E}^{(1)} + \delta^2\mathbf{E}^{(2)} + \dots && (11)\\
\mathbf{B}(\mathbf{r},t) &= \mathbf{B}^{(0)} + \delta\mathbf{B}^{(1)} + \delta^2\mathbf{B}^{(2)} + \dots && (12)\\
\mathbf{J}(\mathbf{r},t) &= \mathbf{J}^{(0)} + \delta\mathbf{J}^{(1)} + \delta^2\mathbf{J}^{(2)} + \dots && (13)
\end{aligned}
$$

At the order-0, the Vlasov equation is:

$$
\frac{\partial f_e^{(0)}}{\partial t_\omega} + \mathbf{v}\cdot\nabla_\lambda f_e^{(0)} + q_e\left(\mathbf{v}\times\mathbf{B}^{(0)}\right)\cdot\frac{\partial f_e^{(0)}}{\partial\mathbf{p}} = 0 \qquad (14)
$$

The equilibrium distribution function does not vary on temporal and spatial length scale of the incident wave, so we can neglect the first two terms and the 0-order Vlasov equations becomes:

$$
q_e\left(\mathbf{v}\times\mathbf{B}^{(0)}\right)\cdot\frac{\partial f_e^{(0)}}{\partial\mathbf{p}} = 0 \qquad (15)
$$

This relation gives us the equilibrium distribution function. The 0-order Maxwell equations are:

$$
\begin{aligned}
\frac{\partial\mathbf{B}^{(0)}}{\partial t_\omega} &= 0 && (16)\\
\nabla_\lambda\times\mathbf{B}^{(0)} &= \mu_0\mathbf{J}^{(0)} && (17)
\end{aligned}
$$

The equation (29) gives the independence of the equilibrium magnetic field by the incident wave time-scale, this field is in general assumed to be costant in time, independent by all the time-scales. $\mathbf{B}^{(0)}$does not depend by the space-scale of the wave, so the left term of the (30) is assumed to be zero and in this case we have $\mathbf{J}^{(0)} = 0$; this is not a general assumption, but this derive from the fact that we are working with the wave ordering, not with the tokamak ordering, infact the equilibrium magnetic field is independent by the wave space-scale but not by the space in general and the equilibrium current is of course not due to the incident wave, using the tokamak ordering we obtain the equation for $\mathbf{B}^{(0)}$ with the spatial derivatives referred to the space-scale of the gyroradius.

At the order-1, the Vlasov equation is:

$$
\frac{\partial f_e^{(1)}}{\partial t_\omega}+\mathbf{v}\cdot\nabla_\lambda f_e^{(1)}+q_e\left(\mathbf{v}\times\mathbf{B}^{(0)}\right)\cdot\frac{\partial f_e^{(1)}}{\partial\mathbf{p}} = -q_e\left(\mathbf{E}^{(1)}+\mathbf{v}\times\mathbf{B}^{(1)}\right)\cdot\frac{\partial f_e^{(0)}}{\partial\mathbf{p}}-\mathbf{v}\cdot\nabla_{\|\widetilde{\lambda}_\|}f_e^{(0)} \qquad (18)
$$

Remembering that the equilibrium distribution function does not depend by the fluctuation space-scale, we can neglect the second left term, and we obtain the linearized Vlasov equation:

$$\frac{\partial f_e^{(1)}}{\partial t_\omega} + \mathbf{v}\cdot\nabla_\lambda f_e^{(1)} + q_e\left(\mathbf{v}\times\mathbf{B}^{(0)}\right)\cdot\frac{\partial f_e^{(1)}}{\partial \mathbf{p}} = -q_e\left(\mathbf{E}^{(1)} + \mathbf{v}\times\mathbf{B}^{(1)}\right)\cdot\frac{\partial f_e^{(0)}}{\partial \mathbf{p}} \quad (19)$$

The order-1 Maxwell equations are:

$$\nabla_\lambda \times \mathbf{E}^{(1)} = -\frac{\partial \mathbf{B}^{(1)}}{\partial t_\omega} \quad (20)$$

$$\nabla_\lambda \times \mathbf{B}^{(1)} + \nabla_{\|\widetilde{\lambda}_\|} \times \mathbf{B}^{(0)} = \varepsilon_0\mu_0\frac{\partial \mathbf{E}^{(1)}}{\partial t_\omega} + \mu_0\mathbf{J}^{(1)} \quad (21)$$

The equilibrium magnetic field is independent by the fluctuations parallel space-scale, so at the order-1 we have the Maxwell equation in closed form:

$$\nabla_\lambda \times \mathbf{E}^{(1)} = -\frac{\partial \mathbf{B}^{(1)}}{\partial t_\omega} \quad (22)$$

$$\nabla_\lambda \times \mathbf{B}^{(1)} = \mu_0\varepsilon_0\frac{\partial \mathbf{E}^{(1)}}{\partial t_\omega} + \mu_0\mathbf{J}^{(1)} \quad (23)$$

And the 1-order current is given by:

$$\mathbf{J}^{(1)} = q_e\int \mathbf{v} f_e^{(1)}\mathrm{d}\mathbf{p} \quad (24)$$

Where $f_e^{(1)}$ is solution of the linearized Vlasov equation (19). At the second order the Vlasov equation becomes:

$$\frac{\partial f_e^{(2)}}{\partial t_\omega} + \frac{\partial f_e^{(0)}}{\partial t_{\widetilde{\omega}}} + \mathbf{v}\cdot\nabla_\lambda f_e^{(2)} + \mathbf{v}\cdot\nabla_{\|\widetilde{\lambda}_\|} f_e^{(1)} + q_e\left(\mathbf{v}\times\mathbf{B}^{(0)}\right)\cdot\frac{\partial f_e^{(2)}}{\partial \mathbf{p}} =$$

$$= -q_e\left(\mathbf{E}^{(2)} + \mathbf{v}\times\mathbf{B}^{(2)}\right)\cdot\frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - q_e\left(\mathbf{E}^{(1)} + \mathbf{v}\times\mathbf{B}^{(1)}\right)\cdot\frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \quad (25)$$

The equilibrium and the first order distribution function do not vary with the time-scale and the space-scale of the fluctuations, tha are very slow respect the incident waves scales, so at order-2 the Vlasov equation is:

$$\frac{\partial f_e^{(2)}}{\partial t_\omega} + \mathbf{v}\cdot\nabla_\lambda f_e^{(2)} + q_e\left(\mathbf{v}\times\mathbf{B}^{(0)}\right)\cdot\frac{\partial f_e^{(2)}}{\partial \mathbf{p}} =$$

$$= -q_e\left(\mathbf{E}^{(2)} + \mathbf{v}\times\mathbf{B}^{(2)}\right)\cdot\frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - q_e\left(\mathbf{E}^{(1)} + \mathbf{v}\times\mathbf{B}^{(1)}\right)\cdot\frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \quad (26)$$

At the order-2 of the Maxwell equations we have:

$$\nabla_\lambda \times \mathbf{E}^{(2)} + \nabla_{\|\widetilde{\lambda}_\|} \times \mathbf{E}^{(1)} = -\frac{\partial \mathbf{B}^{(0)}}{\partial t_{\widetilde{\omega}}} - \frac{\partial \mathbf{B}^{(1)}}{\partial t_\omega} \tag{27}$$

$$\nabla_\lambda \times \mathbf{B}^{(2)} + \nabla_{\|\widetilde{\lambda}_\|} \times \mathbf{B}^{(1)} = \varepsilon_0 \mu_0 \frac{\partial \mathbf{E}^{(2)}}{\partial t_\omega} + \mu_0 \mathbf{J}^{(2)} \tag{28}$$

$\mathbf{B}^{(0)}$ is assumed to be stationary, the first order fields, $\mathbf{E}^{(1)}$ and $\mathbf{B}^{(1)}$, due to the wave fields, vary with the space-scale of the wave, that is faster than the parallel space-scale of the fluctuations, so we can neglect the second left terms of the equations, and we obtain the second order Maxwell equations in closed form:

$$\nabla_\lambda \times \mathbf{E}^{(2)} = -\frac{\partial \mathbf{B}^{(2)}}{\partial t_\omega} \tag{29}$$

$$\nabla_\lambda \times \mathbf{B}^{(2)} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}^{(2)}}{\partial t_\omega} + \mu_0 \mathbf{J}^{(2)} \tag{30}$$

And the 1-order current is given by:

$$\mathbf{J}^{(2)} = q_e \int \mathbf{v} f_e^{(2)} \mathrm{d}\mathbf{p} \tag{31}$$

Where $f_e^{(2)}$ is solution of the second order Vlasov equation (26). So using this kind of expansion, we have obtained at orders 1 and 2 the Maxwell-Vlasov system in closed form (sets of equations (19),(89),(90) and (26),(29),(30)). This two sets of equations are the starting point for the study of the interaction between the incident wave and the fluctuations, that are treated as two indipendent linear waves, solutions of the first order Maxwell-Vlasov equations (19),(89),(90).

2 Three waves mixing

The incident wave and the fluctuations are considered two indipendent linear waves, solution of the first-order Maxwell-Vlasov system. To semplify the notation, we now obmit the pedix notation of the temporal and spatial operators in the first and second order equations, and we imply that the derivatives are referred to the incident wave space and time-scale, $\lambda$and ω. We introduce also the Vlasov linear operator, $\mathcal{L}$:

$$\mathcal{L} f_e = \frac{\partial f_e}{\partial t} + \mathbf{v} \cdot \nabla f_e + q_e \left(\mathbf{v} \times \mathbf{B}^{(0)} \right) \cdot \frac{\partial f_e}{\partial \mathbf{p}} \tag{32}$$

And the force $\mathbf{F}^{(n)}$that the field $\mathbf{E}^{(n)}(\mathbf{r}, t)$ and $\mathbf{B}^{(n)}(\mathbf{r}, t)$, of order (n), exert on an electron of velocity $\mathbf{v}$:

$$\mathbf{F}^{(n)} = q_e \left(\mathbf{E}^{(n)} + \mathbf{v} \times \mathbf{B}^{(n)} \right) \tag{33}$$

Using this compact notation, the first order equation (linearized Vlasov equation (19)), which describes the linear wave propagation, becomes:

$$\mathcal{L} f_e^{(1)} = -\mathbf{F}^{(1)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} \tag{34}$$

The incident wave $\left[f_e^{(1)}, \mathbf{E}^{(1)}, \mathbf{B}^{(1)} \right]$ and the fluctuations $\left[\widetilde{f}_e^{(1)}, \widetilde{\mathbf{E}}^{(1)}, \widetilde{\mathbf{B}}^{(1)} \right]$ are two independent solutions of this linearized Vlasov equation and of the first order Maxwell equations:

$$\nabla \times \mathbf{E}^{(1)} = -\frac{\partial \mathbf{B}^{(1)}}{\partial t} \tag{35}$$

$$\nabla \times \mathbf{B}^{(1)} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}^{(1)}}{\partial t} + \mu_0 \mathbf{J}^{(1)} \tag{36}$$

The system of equations (34), (89),(90) is linear, so the sum of the incident wave and the fluctuation wave is also a solution of this system. At the first order, as we can observe in the equations, there is no coupling between the two waves, that are independent solutions and coexist in the plasma. Using the compact notation, the second order Vlasov equation (26) becomes:

$$\mathcal{L} f_e^{(2)} = -\mathbf{F}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{37}$$

This equation is linear in second order quantities, so we can construct a second order solution a sum of three elements $\left[f_e^{(2)}, \mathbf{E}^{(2)}, \mathbf{B}^{(2)} \right]$, $\left[\widetilde{f}_e^{(2)}, \widetilde{\mathbf{E}}^{(2)}, \widetilde{\mathbf{B}}^{(2)} \right]$ and $\left[\check{f}_e^{(2)}, \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)} \right]$, each one of those is solution of second order Maxwell equations:

$$\nabla \times \mathbf{E}^{(2)} = -\frac{\partial \mathbf{B}^{(2)}}{\partial t} \tag{38}$$

$$\nabla \times \mathbf{B}^{(2)} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}^{(2)}}{\partial t} + \mu_0 \mathbf{J}^{(2)} \tag{39}$$

With the associated second order current given by the expression:

$$\mathbf{J}^{(2)} = q_e \int \mathbf{v} f_e^{(2)} \mathrm{d}\ \mathbf{p} \tag{40}$$

Elements $\left[f_e^{(2)}, \mathbf{E}^{(2)}, \mathbf{B}^{(2)} \right]$ and $\left[\widetilde{f}_e^{(2)}, \widetilde{\mathbf{E}}^{(2)}, \widetilde{\mathbf{B}}^{(2)} \right]$ satisfies the second order Vlasov equation without cross terms in the linear wave quantities (first order quantities):

$$\mathcal{L}f_e^{(2)} = -\mathbf{F}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{41}$$

$$\mathcal{L}\widetilde{f}_e^{(2)} = -\widetilde{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} \tag{42}$$

While the second order Vlasov equation for element $\left[\check{f}_e^{(2)}, \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)}\right]$ includes the cross terms in the linear wave quantities (see appendix C):

$$\mathcal{L}\check{f}_e^{(2)} = -\check{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{43}$$

Elements $\left[f_e^{(2)}, \mathbf{E}^{(2)}, \mathbf{B}^{(2)}\right]$ and $\left[\widetilde{f}_e^{(2)}, \widetilde{\mathbf{E}}^{(2)}, \widetilde{\mathbf{B}}^{(2)}\right]$ are second order corrections to the linear waves $\left[f_e^{(1)}, \mathbf{E}^{(1)}, \mathbf{B}^{(1)}\right]$and $\left[\widetilde{f}_e^{(1)}, \widetilde{\mathbf{E}}^{(1)}, \widetilde{\mathbf{B}}^{(1)}\right]$, while $\left[\check{f}_e^{(2)}, \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)}\right]$ is a new wave resulting from the bilinear interaction between the linear waves[9], remembering that the two original linear waves are respectively incident wave and fluctuations, this new wave is the scattered wave. So it has been clearly shown that scattering by fluctuations is a second order process due to the bilinear interaction between two linear waves: the incident wave and the fluctuations wave. Observing equation (140), we note that the left -hand side term and the first right-hand side term, involving the quantities due to the scattered wave, are identical to the linearized Vlasov equation (34), so these two terms describe the linear propagation of the scattered wave, the second and the third right-hand terms of the equation describe the bilinear interactions between the incident wave and the fluctuations. The electromagnetic field of the scattered wave is then solution of the second-order wave equation (see appendix C):

$$\nabla \times \nabla \times \check{\mathbf{E}}^{(2)} + \varepsilon_0 \mu_0 \frac{\partial^2 \check{\mathbf{E}}^{(2)}}{\partial \mathrm{t}^2} = -\mu_0 \frac{\partial}{\partial \mathrm{t}} \left(\check{\mathbf{J}}_{lin}^{(2)} + \check{\mathbf{J}}_{int}^{(2)}\right) \tag{44}$$

Where $\check{\mathbf{J}}_{lin}^{(2)}$ is the second order current due to the linear response of the plasma to the field of the scattered wave, and $\mathbf{J}_{int}^{(2)}$ is the current due to the interaction of the incident wave with the fluctuations, and is given by:

$$\check{\mathbf{J}}_{int}^{(2)} = q_e \int \mathbf{v} \check{f}_{int}^{(2)} \mathrm{d}\mathbf{p} \tag{45}$$

Where the function $\check{f}_{int}^{(2)}$satisfies the bilinear part of the second order Vlasov equation (see appendix B):

$$\mathcal{L}\check{f}_{int}^{(2)} = -\mathbf{F}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{46}$$

Applying the Fourier analysis to the (144), we obtain the non homogeneous second order wave equation[9]:

$$\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}\left(\check{\mathbf{k}},\check{\omega}\right)=-\frac{i}{\varepsilon_0\check{\omega}}\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) \tag{47}$$

Where $\mathbb{D}$ is the dispersion tensor, that can be expressed in function of the dielectric tensor, $\mathbb{K}$:

$$\mathbb{D}\left(\mathbf{k},\omega\right)=\frac{c^2}{\omega^2}\left(\mathbf{kk}-k^2\right)+\mathbb{K}\left(\mathbf{k},\omega\right) \tag{48}$$

An explicit expression for the current $\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)$, for fluctuations of density, magnetic field, and particle velocitity is obtained by solving equation (142) in the Fourier space and using the result in expression (143)[9] . We have used a kinetic approach for studying the wave-wave interaction, this kinetic approach, based on the perturbative analysis of the Vlasov equation, takes in account the effects of the temperature and it is the most general approach that can be used for studying this kind of problems, until this point we have not make assumptions regarding the characteristics of the plasma.

3 Fluid approach in cold plasma limit

We now study the interaction between the incident wave and the fluctuations in a cold plasma[10, 12]. In the cold plasma limit, the thermal motions of the particle and in general all the temperature effects are neglected, for this reason we can use a fluid approach alternative to the general kinetic approach illustrated in the precedent paragraph. This fluid approach is based on the perturbative analysis of the MHD equations, that can be derived by the moments anaysis of the Vlasov equation (158)[9, 10]. The MHD tractation is the low temperature limit of the kinetic tractation, then making this low temperature limit we derive the general form of the MHD equations, and we specialize this set of equations for the cold plasma case. By means of the perturbative expansion and of the Fourier analysis of the cold-fluid equation we obtain an explicit expression for the interaction current $\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)$, valid in the cold plasma limit.

The zero order and the first order moments of the distribution function are respectively the electron density and the electron flux:

$$n_e = \int f_e \mathrm{d}\mathbf{p} \tag{49}$$

$$\overline{\mathbf{v}} = \int \mathbf{v} f_e \mathrm{d}\mathbf{p} \tag{50}$$

The electron velocity has two components, the fluid velocity $\mathbf{u}$ and the thermal velocity $\mathbf{w}$, due to the temperaure kinetic effects:

$$\mathbf{v}=\mathbf{u}+\mathbf{w} \tag{51}$$

The average of the thermal velocity is zero, the fluid velocity is connected to the electron flux by the relation:

$$\mathbf{u} =< \mathbf{v} >= \frac{1}{n_e} \int \mathbf{v} f_e \mathrm{d}\mathbf{p} \tag{52}$$

So we have $\overline{\mathbf{v}} = n_e \mathbf{u}$. If we calculate the zero, the first and the second order moments of Vlasov equation (158), we obtain[9, 13]:

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \overline{\mathbf{v}} = 0 \tag{53}$$

$$\frac{\partial \overline{\mathbf{v}}}{\partial t} + \nabla \cdot \overline{\mathbf{v}\mathbf{v}} - \frac{q_e}{m_e} \left(n_e \mathbf{E} + \overline{\mathbf{v}} \times \mathbf{B} \right) = 0 \tag{54}$$

$$\frac{\partial \overline{\mathbf{v}\mathbf{v}}}{\partial t} + \nabla \cdot \overline{\mathbf{v}\mathbf{v}\mathbf{v}} - \frac{q_e}{m_e} \left[\overline{\mathbf{v}} \mathbf{E} + \mathbf{E} \overline{\mathbf{v}} + (\overline{\mathbf{v}\mathbf{v}} \times \mathbf{B}) - (\mathbf{B} \times \overline{\mathbf{v}\mathbf{v}}) \right] = 0 \tag{55}$$

Where the diad $\overline{\mathbf{v}\mathbf{v}}$ and the triad $\overline{\mathbf{v}\mathbf{v}\mathbf{v}}$ are respectively the second and the third order moment of f_e:

$$\overline{\mathbf{v}\mathbf{v}} = \int \mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} \tag{56}$$

$$\overline{\mathbf{v}\mathbf{v}\mathbf{v}} = \int \mathbf{v}\mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} \tag{57}$$

The equation (53) is the continuity equation, the (54) is the equation of motion, and (55) is the energy conservation equation, so with the moments analysis of Vlasov equation we have easily derived the MHD equations in a general form. In the limit of low temperature, we can study the wave-fluctuation interaction starting from the perturbative expansion of the set of equations (53), (54), (55), coupled with the Maxwell equations. We can use this approach in the generic low temperature limit, that is not exactly equivalent to the cold plasma limit. Infact the diad $\overline{\mathbf{v}\mathbf{v}}$ imply the stress tensor $\mathbb{P}$ [13]:

$$\overline{\mathbf{v}\mathbf{v}} = \int \mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} = n_e \mathbf{u}\mathbf{u} + 2\mathbf{u}\overline{\mathbf{w}} + \overline{\mathbf{w}\mathbf{w}} = n_e \mathbf{u}\mathbf{u} + \frac{\mathbb{P}}{m_e} \tag{58}$$

$$\mathbb{P} = m_e \int \mathbf{w}\mathbf{w} f_e \mathrm{d}\mathbf{p} = m_e \overline{\mathbf{w}\mathbf{w}} = m_e n_e < \mathbf{w}\mathbf{w} > \tag{59}$$

In the low temperature limit, this tensor is in general not negligible, in this limit is taken in account of the second order thermal correction due to the term $< \mathbf{w}\mathbf{w} >$, this is the reason because we generally need to include the energy conservation equation (55) to have a closed set of equations at the orders 0, 1 , and 2. Remembering that $\overline{\mathbf{v}} = n_e \mathbf{u}$ and using the expression (58), the continuity (53) and the motion equation (54) become:

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e \mathbf{u}) = 0 \tag{60}$$

$$m_e \frac{\partial}{\partial t}\left(n_e \mathbf{u}\right) + m_e \nabla \cdot \left(n_e \mathbf{u}\mathbf{u}\right) = q_e n_e \left(\mathbf{E} + \mathbf{u} \times \mathbf{B}\right) - \nabla \cdot \mathbb{P} \tag{61}$$

If we use the relations:

$$\frac{\partial}{\partial t}\left(n_e \mathbf{u}\right) = \frac{\partial n_e}{\partial t}\mathbf{u} + n_e \frac{\partial \mathbf{u}}{\partial t} \tag{62}$$

$$\nabla \cdot \left(n_e \mathbf{u}\mathbf{u}\right) = \mathbf{u}\nabla \cdot \left(n_e \mathbf{u}\right) + n_e \left(\mathbf{u} \cdot \nabla\right)\mathbf{u} \tag{63}$$

And the contituity equation, the equation for motion finally becomes:

$$m_e n_e \left[\frac{\partial \mathbf{u}}{\partial t} + \left(\mathbf{u} \cdot \nabla\right)\mathbf{u}\right] = e n_e \left(\mathbf{E} + \mathbf{u} \times \mathbf{B}\right) - \nabla \cdot \mathbb{P} \tag{64}$$

So we can see from this explicit form of the equation that if the stress tensor is not negligible, we need to derive an equation for the evolution of this tensor starting by the implicit conservation energy equation (55)[13]. In the cold plasma limit, all the thermal effects, also the second order corrections due to $< \mathbf{ww} >$are neglected, so we can neglect $\mathbb{P}$ and the continuity and motion equations are:

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \left(n_e \mathbf{u}\right) = 0 \tag{65}$$

$$m_e n_e \left[\frac{\partial \mathbf{u}}{\partial t} + \left(\mathbf{u} \cdot \nabla\right)\mathbf{u}\right] = e n_e \left(\mathbf{E} + \mathbf{u} \times \mathbf{B}\right) \tag{66}$$

As we can observe, these two equations, coupled with the Maxwell equations, form a closed system of equations, so for studying the interaction between the incident wave and the fluctuations in the cold plasma limit we can start by this system of equations and we don't need to insert the energy conservation equation. Also in the cold approximation, we separe the dynamics of the incident wave from the dynamics of the fluctuations as we have done in the general kinetic approach:

$$\nabla = \nabla_\lambda + \delta \nabla_{\|\widetilde{\lambda}_\|} \tag{67}$$

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial t_\omega} + \delta^2 \frac{\partial}{\partial t_{\widetilde{\omega}}} \tag{68}$$

Where δ is the same ordering (4) parameter used for the kinetic tractation. Using this parameter, we expand the physical quantities, assuming $\mathbf{u}^{(0)} = 0$ and $\mathbf{E}^{(0)} = 0$ (at the equilibrium the electric field and the electron fluid velocity, that is the average velocity of the electrons, are zero):

$$n_e\left(\mathbf{r}, t\right) = n_e^{(0)} + \delta n_e^{(1)} + \delta^2 n_e^{(2)} + \ldots \tag{69}$$

$$\mathbf{u}\left(\mathbf{r}, t\right) = \delta \mathbf{u}^{(1)} + \delta^2 \mathbf{u}^{(2)} + \ldots \tag{70}$$

$$\mathbf{E}\left(\mathbf{r}, t\right) = \delta \mathbf{E}^{(1)} + \delta^2 \mathbf{E}^{(2)} + \ldots \tag{71}$$

$$\mathbf{B}\left(\mathbf{r}, t\right) = \mathbf{B}^{(0)} + \delta \mathbf{B}^{(2)} + \delta^2 \mathbf{B}^{(2)} + \ldots \tag{72}$$

The zero, the first and the second order of the continuity equation are:

$$\frac{\partial n_e^{(0)}}{\partial t_\omega} = 0 \tag{73}$$

$$\frac{\partial n_e^{(1)}}{\partial t_\omega} + \nabla_\lambda \cdot \left(n_e^{(0)} \mathbf{u}^{(1)} \right) = 0 \tag{74}$$

$$\frac{\partial n_e^{(2)}}{\partial t_\omega} + \frac{\partial n_e^{(0)}}{\partial t_{\widetilde{\omega}}} + \nabla_\lambda \cdot \left(n_e^{(0)} \mathbf{u}^{(2)} + n_e^{(1)} \mathbf{u}^{(1)} \right) + \nabla_{\|\widetilde{\lambda}_\|} \cdot \left(n_e^{(0)} \mathbf{u}^{(1)} \right) = 0 \tag{75}$$

From the equation (73) we can say that the equilibrium density is indepndent by the incident wave time-scale, $n_e^{(0)}$ is in general assumed to be costant in time, independent by all the time-scales. The zero order equilibrium quantities, and the first order quantities due to the wave fields, do not vary with the parallel space-scale of the fluctuations, that is too slow, so we can neglect the last right term of the (96), so the first and the second order contituity equation become:

$$\frac{\partial n_e^{(1)}}{\partial t_\omega} + \nabla_\lambda \cdot \left(n_e^{(0)} \mathbf{u}^{(1)} \right) = 0 \tag{76}$$

$$\frac{\partial n_e^{(2)}}{\partial t_\omega} + \nabla_\lambda \cdot \left(n_e^{(0)} \mathbf{u}^{(2)} + n_e^{(1)} \mathbf{u}^{(1)} \right) = 0 \tag{77}$$

At the first and second order the equation of motion is:

$$m_e n_e^{(0)} \frac{\partial \mathbf{u}^{(1)}}{\partial t_\omega} = q_e n_e^{(0)} \left(\mathbf{E}^{(1)} + \mathbf{u}^{(1)} \times \mathbf{B}^{(0)} \right) \tag{78}$$

$$m_e n_e \frac{\partial \mathbf{u}^{(2)}}{\partial t_\omega} + m_e n_e^{(1)} \frac{\partial \mathbf{u}^{(1)}}{\partial t_\omega} + \left(\mathbf{u}^{(1)} \cdot \nabla_\lambda \right) \mathbf{u}^{(1)} =$$

$$= e n_e^{(1)} \left(\mathbf{E}^{(1)} + \mathbf{u}^{(1)} \times \mathbf{B}^{(0)} \right) + e n_e^{(0)} \left(\mathbf{E}^{(2)} + \mathbf{u}^{(1)} \times \mathbf{B}^{(1)} + \mathbf{u}^{(2)} \times \mathbf{B}^{(0)} \right) \tag{79}$$

Dividing these equations by $n_e m_e$ and recovering the (165) in the second left and in the first right term of the (97), we obtain:

$$\frac{\partial \mathbf{u}^{(1)}}{\partial t_\omega} = \frac{q_e}{m_e} \left(\mathbf{E}^{(1)} + \mathbf{u}^{(1)} \times \mathbf{B}^{(0)} \right) \tag{80}$$

$$\frac{\partial \mathbf{u}^{(2)}}{\partial t_\omega} + \left(\mathbf{u}^{(1)} \cdot \nabla_\lambda \right) \mathbf{u}^{(1)} = \frac{q_e}{m_e} \left(\mathbf{E}^{(2)} + \mathbf{u}^{(2)} \times \mathbf{B}^{(0)} \right) + \frac{q_e}{m_e} \left(\mathbf{u}^{(1)} \times \mathbf{B}^{(1)} \right) \tag{81}$$

The equation (165) is the linearized equation of motion. We now obmit the pedix notation of the temporal and spatial operators and we imply that the

derivatives are referred to the incident wave space and time-scale, $\lambda$and ω. The first order equations are then:

$$\frac{\partial n_e^{(1)}}{\partial t} + \nabla \cdot \left(n_e^{(0)} \mathbf{u}^{(1)} \right) = 0 \tag{82}$$

$$\frac{\partial \mathbf{u}^{(1)}}{\partial t} = \frac{q_e}{m_e} \left(\mathbf{E}^{(1)} + \mathbf{u}^{(1)} \times \mathbf{B}^{(0)} \right) \tag{83}$$

This equation are coupled with the first order Maxwell equations:

$$\nabla \times \mathbf{E}^{(1)} = -\frac{\partial \mathbf{B}^{(1)}}{\partial t} \tag{84}$$

$$\nabla \times \mathbf{B}^{(1)} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}^{(1)}}{\partial t} + \mu_0 \mathbf{J}^{(1)} \tag{85}$$

Where the first order current is:

$$\mathbf{J}^{(1)} = q_e n_e^{(0)} \mathbf{u}^{(1)} \tag{86}$$

As in the kinetic tractation, at the first order we have obtained a system of linear equation in closed form, the incident wave $\left[n_e^{(1)}, \mathbf{u}^{(1)}, \mathbf{E}^{(1)}, \mathbf{B}^{(1)} \right]$and the fluctuations $\left[\widetilde{n}_e^{(1)}, \widetilde{\mathbf{u}}^{(1)}, \widetilde{\mathbf{E}}^{(1)}, \widetilde{\mathbf{B}}^{(1)} \right]$ are two independent solutions of this linear set of equation, and also the sum of the two waves is a solution. The analogy with the general kinetic approach is total, and also in this cold fluid approximation we can observe that at the first order there is no coupling between the incident wave and thec fluctuation, that are independent modes and coexist in the plasma. The second order cold-fluid equations are:

$$\frac{\partial n_e^{(2)}}{\partial t} + \nabla \cdot \left(n_e^{(0)} \mathbf{u}^{(2)} + n_e^{(1)} \mathbf{u}^{(1)} \right) = 0 \tag{87}$$

$$\frac{\partial \mathbf{u}^{(2)}}{\partial t} + \left(\mathbf{u}^{(1)} \cdot \nabla \right) \mathbf{u}^{(1)} = \frac{q_e}{m_e} \left(\mathbf{E}^{(2)} + \mathbf{u}^{(2)} \times \mathbf{B}^{(0)} \right) + \frac{q_e}{m_e} \left(\mathbf{u}^{(1)} \times \mathbf{B}^{(1)} \right) \tag{88}$$

This equations are linear in second order quantities, so also in this cold-fluid approach we can construct a second order solution a sum of three elements $\left[n_e^{(2)}, \mathbf{u}^{(2)}, \mathbf{E}^{(2)}, \mathbf{B}^{(2)} \right]$, $\left[\widetilde{n}_e^{(2)}, \widetilde{\mathbf{u}}^{(2)}, \widetilde{\mathbf{E}}^{(2)}, \widetilde{\mathbf{B}}^{(2)} \right]$ and $\left[\check{n}_e^{(2)}, \check{\mathbf{u}}^{(2)} \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)} \right]$, with the second order electric and magnetic fields solutions of second order Maxwell equations:

$$\nabla \times \mathbf{E}^{(2)} = -\frac{\partial \mathbf{B}^{(2)}}{\partial t} \tag{89}$$

$$\nabla \times \mathbf{B}^{(2)} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}^{(2)}}{\partial t} + \mu_0 \mathbf{J}^{(2)} \tag{90}$$

And the second order current given by the expression:

$$\mathbf{J}^{(2)} = q_e n_e^{(0)} \mathbf{u}^{(2)} + q_e n_e^{(1)} \mathbf{u}^{(1)} \tag{91}$$

The second order quantities due to the incident wave and to the fluctuations satisfies the second order fluid equations without cross term in first order quantities:

$$\frac{\partial n_e^{(2)}}{\partial t} + \nabla \cdot \left(n_e^{(0)} \mathbf{u}^{(2)} + n_e^{(1)} \mathbf{u}^{(1)} \right) = 0 \tag{92}$$

$$\frac{\partial \mathbf{u}^{(2)}}{\partial t} + \left(\mathbf{u}^{(1)} \cdot \nabla \right) \mathbf{u}^{(1)} = \frac{q_e}{m_e} \left(\mathbf{E}^{(2)} + \mathbf{u}^{(2)} \times \mathbf{B}^{(0)} \right) + \frac{q_e}{m_e} \left(\mathbf{u}^{(1)} \times \mathbf{B}^{(1)} \right) \tag{93}$$

$$\frac{\partial \widetilde{n}_e^{(2)}}{\partial t} + \nabla \cdot \left(n_e^{(0)} \mathbf{u}^{(2)} + \widetilde{n}_e^{(1)} \widetilde{\mathbf{u}}^{(1)} \right) = 0 \tag{94}$$

$$\frac{\partial \widetilde{\mathbf{u}}^{(2)}}{\partial t} + \left(\widetilde{\mathbf{u}}^{(1)} \cdot \nabla \right) \widetilde{\mathbf{u}}^{(1)} = \frac{q_e}{m_e} \left(\widetilde{\mathbf{E}}^{(2)} + \widetilde{\mathbf{u}}^{(2)} \times \widetilde{\mathbf{B}}^{(0)} \right) + \frac{q_e}{m_e} \left(\widetilde{\mathbf{u}}^{(1)} \times \widetilde{\mathbf{B}}^{(1)} \right) \tag{95}$$

While for the second order field $\left[\check{n}_e^{(2)}, \check{\mathbf{u}}^{(2)} \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)} \right]$, the equations become:

$$\frac{\partial \check{n}_e^{(2)}}{\partial t} + \nabla \cdot \left(n_e^{(0)} \check{\mathbf{u}}^{(2)} + \widetilde{n}_e^{(1)} \mathbf{u}^{(1)} + n_e^{(1)} \widetilde{\mathbf{u}}^{(1)} \right) = 0 \tag{96}$$

$$\frac{\partial \check{\mathbf{u}}}{\partial t} + \left(\widetilde{\mathbf{u}}^{(1)} \cdot \nabla \right) \mathbf{u}^{(1)} + \left(\mathbf{u}^{(1)} \cdot \nabla \right) \widetilde{\mathbf{u}}^{(1)} =$$

$$= \frac{q_e}{m_e} \left(\check{\mathbf{E}}^{(2)} + \check{\mathbf{u}}^{(2)} \times \check{\mathbf{B}}^{(0)} \right) + \frac{q_e}{m_e} \left(\widetilde{\mathbf{u}}^{(1)} \times \mathbf{B}^{(1)} + \mathbf{u}^{(1)} \times \widetilde{\mathbf{B}}^{(1)} \right) \tag{97}$$

From the equations (96) and (97) we can deduce that the wave $\left[\check{n}_e^{(2)}, \check{\mathbf{u}}^{(2)} \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)} \right]$ is generated by the interaction between the first order linear fields due the incident wave and the first order field due to the fluctuations, so $\left[\check{n}_e^{(2)}, \check{\mathbf{u}}^{(2)} \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)} \right]$ is the scattered wave and also in the cold fluid approximation scattering by fluctuations is a second order process, we have then shown the total analogy between the general kinetic tractation and the cold-fluid approximation. Then also in the cold plasma limit the electromagnetic field is solution of the second-order wave equation:

$$\nabla \times \nabla \times \check{\mathbf{E}}^{(2)} + \varepsilon_0 \mu_0 \frac{\partial^2 \check{\mathbf{E}}^{(2)}}{\partial \mathrm{t}^2} = -\mu_0 \frac{\partial}{\partial \mathrm{t}} \left(\check{\mathbf{J}}_{lin}^{(2)} + \check{\mathbf{J}}_{int}^{(2)} \right) \tag{98}$$

Where $\check{\mathbf{J}}_{lin}^{(2)}$ is the second order current due to the linear response of the plasma to the field of the scattered wave, and $\mathbf{J}_{int}^{(2)}$ is the current due to the

interaction of the incident wave with the fluctuations, in the Fourier space the equation becomes:

$$\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}\left(\check{\mathbf{k}},\check{\omega}\right)=-\frac{i}{\varepsilon_0\check{\omega}}\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) \tag{99}$$

The total second order current for the scattered wave is given by the expression[10]:

$$\check{\mathbf{J}}^{(2)}=\check{\mathbf{J}}_{lin}^{(2)}+\check{\mathbf{J}}_{int}^{(2)}=q_e n_e^{(0)}\check{\mathbf{u}}^{(2)}+q_e\widetilde{n}_e^{(1)}\mathbf{u}^{(1)}+q_e n_e^{(1)}\widetilde{\mathbf{u}}^{(1)} \tag{100}$$

So to derive an explicit expression for $\check{\mathbf{J}}^{(2)}$, $\check{\mathbf{J}}_{lin}^{(2)}$ and $\check{\mathbf{J}}_{int}^{(2)}$ we need to obtain $\check{\mathbf{u}}^{(2)}$ solving the equation for motion (97). This equation in the Fourier space becomes:

$$-i\check{\omega}\check{\mathbf{u}}^{(2)}-\frac{q_e B^{(0)}}{m_e}\left(\check{\mathbf{u}}^{(2)}\times\frac{\mathbf{B}^{(0)}}{B^{(0)}}\right)=$$

$$=\frac{q_e}{m_e}\left(\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}+\mathbf{u}^{(1)}\times\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}+\widetilde{\mathbf{u}}^{(1)}\times\mathbf{B}_{\mathbf{k}}^{(1)}\right)-i\left[\left(\mathbf{u}\cdot\widetilde{\mathbf{k}}\right)\widetilde{\mathbf{u}}+\left(\widetilde{\mathbf{u}}\cdot\mathbf{k}\right)\mathbf{u}\right] \tag{101}$$

So we can write:

$$\mathbb{T}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{u}}^{(2)}=\frac{q_e}{m_e}\left(\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}+\mathbf{u}^{(1)}\times\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}+\widetilde{\mathbf{u}}^{(1)}\times\mathbf{B}_{\mathbf{k}}^{(1)}\right)-i\left[\left(\mathbf{u}\cdot\widetilde{\mathbf{k}}\right)\widetilde{\mathbf{u}}+\left(\widetilde{\mathbf{u}}\cdot\mathbf{k}\right)\mathbf{u}\right] \tag{102}$$

Where $\mathbb{T}\left(\check{\mathbf{k}},\check{\omega}\right)$ is the tensor:

$$\mathbb{T}\left(\check{\mathbf{k}},\check{\omega}\right)=-i\check{\omega}\mathbb{I}-\omega_{ce}\left(\underline{\underline{\varepsilon}}^{LC}\cdot\mathbf{b}^{(0)}\right) \tag{103}$$

This tensor is related to the susceptibility tensor $\mathbb{X}$ by the relation[9](see appendix E):

$$\mathbb{T}^{-1}\left(\check{\mathbf{k}},\check{\omega}\right)=-i\frac{\check{\omega}}{\omega_{pe}^2}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right) \tag{104}$$

Using this expression in the (102) we obtain:

$$\widetilde{\mathbf{u}}^{(2)}=-i\frac{e\check{\omega}}{m_e\omega_{pe}^2}\left[\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}+\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left(\mathbf{u}^{(1)}\times\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}+\widetilde{\mathbf{u}}^{(1)}\times\mathbf{B}_{\mathbf{k}}^{(1)}\right)\right]-$$

$$-\frac{\check{\omega}}{\omega_{pe}^2}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left[\left(\mathbf{u}^{(1)}\cdot\widetilde{\mathbf{k}}\right)\widetilde{\mathbf{u}}^{(1)}+\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{k}\right)\mathbf{u}^{(1)}\right] \tag{105}$$

We have obtained an explicit expression for $\check{\mathbf{u}}^{(2)}$, if we insert this expression into the (100), remembering that $\omega_{pe}^2=n_e^{(0)}q_e^2/\varepsilon_0 m_e$, the second order current becomes:

$$\check{\mathbf{J}}^{(2)}=-i\varepsilon_0\check{\omega}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}-i\varepsilon_0\check{\omega}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left(\mathbf{u}^{(1)}\times\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}+\widetilde{\mathbf{u}}\times\mathbf{B}_{\mathbf{k}}^{(1)}\right)-$$

$$-\frac{\varepsilon_0 m_e \check{\omega}}{q_e}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left[\left(\mathbf{u}^{(1)}\cdot\widetilde{\mathbf{k}}\right)\widetilde{\mathbf{u}}^{(1)}+\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{k}\right)\mathbf{u}^{(1)}\right]+q_e n_e^{(1)}\widetilde{\mathbf{u}}^{(1)}+q_e\widetilde{n}_e^{(1)}\mathbf{u}^{(1)} \tag{106}$$

The first term is the component due to the linear response of the plasma to the field of the scattered wave:

$$\check{\mathbf{J}}_{lin}^{(2)}\left(\check{\mathbf{k}},\check{\omega}\right)=-i\varepsilon_0\check{\omega}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}\left(\check{\mathbf{k}},\omega\right)=\mathbb{S}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}^{(2)}\left(\check{\mathbf{k}},\check{\omega}\right) \tag{107}$$

So the current due to the interaction between the incident wave and the fluctiations, in the cold plasma limit is given by the expression:

$$\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)=-i\varepsilon_0\check{\omega}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left(\mathbf{u}^{(1)}\times\widetilde{\mathbf{B}}_{\check{\mathbf{k}}}^{(1)}+\widetilde{\mathbf{u}}\times\mathbf{B}_{\mathbf{k}}^{(1)}\right)-$$

$$-\frac{\varepsilon_0 m_e \check{\omega}}{q_e}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left[\left(\mathbf{u}^{(1)}\cdot\widetilde{\mathbf{k}}\right)\widetilde{\mathbf{u}}^{(1)}+\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{k}\right)\mathbf{u}^{(1)}\right]+q_e n_e^{(1)}\widetilde{\mathbf{u}}^{(1)}+q_e\widetilde{n}_e^{(1)}\mathbf{u}^{(1)} \tag{108}$$

We now exprime the quantities $n_e^{(1)}$, $\mathbf{u}^{(1)}$ and $\mathbf{B}_{\mathbf{k}}^{(1)}$, relatives to the incident wave, in function of the electric field $\mathbf{E}_{\mathbf{k}}^{(1)}$. The first order motion(93) and continuity (92) equations for the incident wave in the Fourier space are given by the expressions (see appendix E):

$$\mathbb{T}\left(\mathbf{k},\omega\right)\cdot\mathbf{u}^{(1)}=\frac{q_e}{m_e}\mathbf{E}_{\mathbf{k}}^{(1)}\left(\mathbf{k},\omega\right) \tag{109}$$

$$\omega n_e^{(1)}-n_e^{(0)}\left(\mathbf{k}\cdot\mathbf{u}^{(1)}\right)=0 \tag{110}$$

Where $\mathbb{T}\left(\mathbf{k},\omega\right)$ is given by the (168), so we derive (see appendix E):

$$\mathbf{u}^{(1)} = -i \quad \frac{\omega\varepsilon_0}{n_e^{(0)}q_e}\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\left(\mathbf{k},\omega\right) \tag{111}$$

$$n_e^{(1)} = -i\frac{\varepsilon_0}{q_e}\left(\mathbf{k}\cdot\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\left(\mathbf{k},\omega\right)\right) \tag{112}$$

From the equation (84) in the Fourier space we derive $\mathbf{B}_{\mathbf{k}}^{(1)}$:

$$\mathbf{B}_{\mathbf{k}}^{(1)}\left(\mathbf{k},\omega\right)=\frac{1}{\omega}\left(\mathbf{k}\times\mathbf{E}_{\mathbf{k}}^{(1)}\left(\mathbf{k},\omega\right)\right)$$

So using the vectorial identities we obtain:

$$\widetilde{\mathbf{u}}^{(1)}\times\mathbf{B}_{\mathbf{k}}^{(1)}=\frac{1}{\omega}\left[\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\mathbf{k}-\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{k}\right)\mathbf{E}_{\mathbf{k}}^{(1)}\right] \tag{113}$$

If we insert the expressions (111), (112) and (113) into the (114) , remembering that $\omega_{ce}=q_eB^{(0)}/m_e$, we finally obtain[9, 12]:

$$\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)=-i\varepsilon_0\omega\left\{\frac{\widetilde{n}_e^{(1)}}{n_e^{(0)}}\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)-i\frac{\omega_{ce}\check{\omega}}{\omega_{pe}^2}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left[\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\times\frac{\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}}{B^{(0)}}\right]\right\}$$

$$-i\varepsilon_0\left\{\frac{\check{\omega}}{\omega}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left[\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\mathbf{k}-\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{k}\right)\mathbf{E}_{\mathbf{k}}^{(1)}-\frac{\omega^2}{\omega_{pe}^2}\left(\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\cdot\widetilde{\mathbf{k}}\right)\widetilde{\mathbf{u}}^{(1)}-\right.\right.$$

$$\left.\left.-\frac{\omega^2}{\omega_{pe}^2}\left(\widetilde{\mathbf{u}}^{(1)}\cdot\mathbf{k}\right)\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\right]+\left(\mathbf{k}\cdot\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\widetilde{\mathbf{u}}^{(1)}\right\}\qquad(114)$$

So we have derived an explicit expression for the current due to the interaction between the incident wave and the fluctuations in the cold plasma limit, so we can insert this expression in the right side of the second-order non-homogeneous wave equation (99) for studying the effects of the fluctuations on the wave propagation and absorption in the cold plasma approximation. We want to study the effects of the density and magnetic field fluctuations, not of the velocity fluctuations, so in the expression for $\check{\mathbf{J}}_{int}^{(2)}$ we obmit the terms due to $\widetilde{\mathbf{u}}^{(1)}$:

$$\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)=-i\varepsilon_0\omega\left\{\frac{\widetilde{n}_e^{(1)}}{n_e^{(0)}}\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)-i\frac{\omega_{ce}\check{\omega}}{\omega_{pe}^2}\mathbb{X}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\left[\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\times\frac{\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}}{B^{(0)}}\right]\right\}\qquad(115)$$

Remembering that $\mathbb{X}=\mathbb{K}-\mathbb{I}$ and that:

$$\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)\times\frac{\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}}{B^{(0)}}=\left(\underset{\equiv}{\varepsilon}^{LC}\cdot\frac{\mathbf{B}_{\mathbf{k}}^{(1)}}{B^{(0)}}\right)\cdot\left(\mathbb{X}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\right)$$

We obtain the interaction current in function of the dielectric tensor and of the incident electric field [2]:

$$\check{\mathbf{J}}_{int}^{(2)}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)=i\varepsilon_0\omega\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\widetilde{n}_e^{(1)}}{n_e^{(0)}}+\right.$$

$$\left.i\frac{\omega_{ce}\check{\omega}}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\check{\omega}\right)\right)\cdot\left(\underset{\equiv}{\varepsilon}^{LC}\cdot\frac{\mathbf{B}_{\mathbf{k}}^{(1)}}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}^{(1)}\qquad(116)$$

4 Mode coupling equation

We now insert the expression (116) in the non-homogeneous wave equation, from this moment we work only with the second order equation, so we obmit the explicit notation, because we imply it:

$$\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)=-\frac{i}{\varepsilon_0\check{\omega}}\check{\mathbf{J}}_{int}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) \tag{117}$$

For the fluctuations we introduce the notation: $\delta n_e\left(\widetilde{\mathbf{k}},\widetilde{\omega}\right)=\widetilde{n}_e^{(1)}$ and $\delta\mathbf{B}\left(\widetilde{\mathbf{k}},\widetilde{\omega}\right)=\widetilde{\mathbf{B}}_{\widetilde{\mathbf{k}}}^{(1)}$ .From the conservation of momentum and energy, the wave vector and the frequency of the scattered wave are given by the expressions [2]:

$$\check{\mathbf{k}}=\mathbf{k}+\widetilde{\mathbf{k}},\quad\check{\omega}=\omega+\widetilde{\omega} \tag{118}$$

So we have:

$$\widetilde{\mathbf{k}}=\check{\mathbf{k}}-\mathbf{k},\quad\widetilde{\omega}=\check{\omega}-\omega \tag{119}$$

Considering several incident modes with wave vector $\mathbf{k}$ and frequency ω, the expression (116) becomes:

$$\check{\mathbf{J}}_{int}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)=i\varepsilon_0\sum_{\mathbf{k},\omega}\omega\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n_e\left(\check{\mathbf{k}}-\mathbf{k},\check{\omega}-\omega\right)}{n_e^{(0)}}+\right.$$

$$\left.+i\frac{\omega_{ce}\check{\omega}}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\check{\omega}\right)\right)\cdot\left(\underline{\underline{\underline{\varepsilon}}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k},\check{\omega}-\omega\right)}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{120}$$

Where $\mathbf{E}_{\mathbf{k}}=\mathbf{E}_{\mathbf{k}}^{(1)}$, $\underline{\underline{\underline{\varepsilon}}}^{LC}$ is the Levi-Civita tensor of order 3, and $\mathbb{K}$ is the cold plasma permittivity tensor. Using this expression for the current due to the fluctuations, the wave equation becomes:

$$\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\check{\omega}\right)=\sum_{\mathbf{k},\omega}\left[\frac{\omega}{\check{\omega}}\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n_e\left(\check{\mathbf{k}}-\mathbf{k},\check{\omega}-\omega\right)}{n_e^{(0)}}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\check{\omega}\right)\right)\cdot\left(\underline{\underline{\underline{\varepsilon}}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k},\check{\omega}-\omega\right)}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{121}$$

Fluctuations are assumed to be slow, so we have $\widetilde{\omega}\ll\omega$ and $\check{\omega}\approx\omega$, as a consequence:

$$\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)=\sum_{\mathbf{k},\omega}\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n_e\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n_e}+\right.$$

$$+i\frac{\omega_{ce}\omega}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\check{\omega}\right)\right)\cdot\left(\underline{\underline{\underline{\varepsilon}}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\Bigg]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{122}$$

Now, we assume that fluctuations of the density and of the magnetic field are sufficienly small to be treated with a perturbative approach, it means that the effects of the fluctuations may be rapresented by a small variation from the normal mode propagation that we can easily study by solving the homogeneous wave equation. In the weak damping approximation, the frequency of the scattered wave is given by the expression [1, 2]:

$$\check{\omega}=\check{\omega}_{\check{k}}+i\frac{\partial}{\partial \mathrm{t}}\qquad\text{with}\ \left|\frac{\partial}{\partial \mathrm{t}}\right|\ll\check{\omega}_{\check{k}} \tag{123}$$

Where $\check{\omega}_{\check{k}}$ is the frequency of the normal mode, that is decribed by the homogeneous equation:

$$\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)=0 \tag{124}$$

$\check{\omega}_{\check{k}}$ can be easily determined by solving the dispersion relation associated with the homogeneous equation. Inserting the expression for $\check{\omega}$ in the wave equation, and expressing the electric field in function of the polarization vector $\mathbf{E}(\mathbf{k})=\mathbf{e}\left(\mathbf{k}\right)E\left(\mathbf{k}\right)$, we obtain:

$$\left[\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)+i\frac{\partial\mathbb{D}}{\partial\check{\omega}}\frac{\partial}{\partial \mathrm{t}}\right]\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\check{E}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\right)=\sum_{\mathbf{k}}\Bigg[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\frac{\delta n_e\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n_e^{(0)}}+$$

$$+i\frac{\omega_{ce}\omega}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\underline{\underline{\underline{\varepsilon}}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\Bigg]\cdot\mathbf{e}\left(\mathbf{k}\right)E_{\mathbf{k}}\left(\mathbf{k}\right) \tag{125}$$

In this equation all the quantities are considered for $\omega=\check{\omega}_{\check{k}}$. Multiplying this equation for $\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)$ and remembering that $\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)=0$, the equation finally becomes:

$$i\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\frac{\partial\mathbb{D}}{\partial\check{\omega}}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\frac{\partial\check{E}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\right)}{\partial \mathrm{t}}=\sum_{\mathbf{k}}\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\Bigg[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\frac{\delta n_e\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n_e}+$$

$$+i\frac{\omega_{ce}\omega}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\underline{\underline{\underline{\varepsilon}}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\Bigg]\cdot\mathbf{e}\left(\mathbf{k}\right)E_{\mathbf{k}}\left(\mathbf{k}\right) \tag{126}$$

In order to study the coupling between the incident and the scattered wave, we introduce the wave energy density:

$$W\left(\mathbf{k}\right)=\varepsilon_{0}\frac{\check{\omega}_{\check{k}}}{2}\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\frac{\partial\mathbb{D}}{\partial\omega}\cdot\mathbf{e}\left(\mathbf{k}\right)\left|\mathbf{E}\left(\mathbf{k}\right)\right|^{2}=\varepsilon_{0}\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)E\left(\mathbf{k}\right)^{2} \tag{127}$$

Where $\mathbb{M}=\frac{\check{\omega}_{\check{k}}}{2}\frac{\partial\mathbb{D}}{\partial\omega}$. The wave amplitude is given by:

$$C\left(\mathbf{k}\right)=W\left(\mathbf{k}\right)^{\frac{1}{2}} \tag{128}$$

So we exprime the electric field in function of the wave amplitude, and we obtain the mode coupling equation:

$$i\frac{\partial C\left(\check{\mathbf{k}}\right)}{\partial\mathrm{t}}=\sum_{\mathbf{k}}\frac{\check{\omega}_{\check{k}}}{2}\left[\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}\frac{\delta n_{e}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n_{e}}+\right.$$

$$\left.+i\frac{\omega_{ce}\check{\omega}_{\check{k}}}{\omega_{pe}^{2}}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\underline{\underline{\underline{\varepsilon}}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}\right]C\left(\mathbf{k}\right) \tag{129}$$

This equation can be written in the form:

$$i\frac{\partial C\left(\check{\mathbf{k}}\right)}{\partial\mathrm{t}}=\sum_{\mathbf{k}}\left[V^{n}\left(\mathbf{k},\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)\frac{\delta n_{e}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n_{e}}+\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}\right]C\left(\mathbf{k}\right) \tag{130}$$

Where $V^{n}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)$and $\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)$ are the coupling coefficients:

$$V^{n}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{\check{\omega}_{\check{k}}}{2}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{131}$$

$$\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{i\omega_{ce}\check{\omega}_{\check{k}}^{2}}{2\omega_{pe}^{2}}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\underline{\underline{\underline{\varepsilon}}}^{LC}\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{132}$$

Using the component notation:

$$V^{n}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{\check{\omega}_{\check{k}}}{2}\frac{\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)\left(\delta_{ij}-K_{ij}\left(\mathbf{k}\right)\right)e_{j}\left(\mathbf{k}\right)}{\left(\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)M_{ij}\check{e}_{j}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(e_{k}^{\star}\left(\mathbf{k}\right)M_{kl}e_{l}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{133}$$

$$V_{m}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{i\omega_{ce}\check{\omega}_{\check{k}}^{2}}{2\omega_{pe}^{2}}\frac{\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)\left(\delta_{il}-K_{il}\left(\check{\mathbf{k}}\right)\right)\epsilon_{lkm}\left(\delta_{kj}-K_{kj}\left(\mathbf{k}\right)\right)e_{j}\left(\mathbf{k}\right)}{\left(\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)M_{ij}\check{e}_{j}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(e_{k}^{\star}\left(\mathbf{k}\right)M_{kl}e_{l}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{134}$$

Appendix A

Appendix B

Appendix C

We now derive the second order Vlasov equation for the wave $\left[\check{f}_e^{(2)}, \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)}\right]$, starting from the second order Vlasov equation:

$$\mathcal{L} f_e^{(2)} = -\mathbf{F}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{135}$$

This equation is linear in second order quantities, so we can construct a second order solution a sum of three elements $\left[f_e^{(2)}, \mathbf{E}^{(2)}, \mathbf{B}^{(2)}\right]$, $\left[\widetilde{f}_e^{(2)}, \widetilde{\mathbf{E}}^{(2)}, \widetilde{\mathbf{B}}^{(2)}\right]$ and $\left[\check{f}_e^{(2)}, \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)}\right]$, where $\left[\check{f}_e^{(2)}, \check{\mathbf{E}}^{(2)}, \check{\mathbf{B}}^{(2)}\right]$ is a purely second order process:

$$\mathcal{L}\left(\check{f}_e^{(2)} + f_e^{(2)} + \widetilde{f}_e^{(2)}\right) = -\left(\check{\mathbf{F}}^{(2)} + \mathbf{F}^{(2)} + \widetilde{\mathbf{F}}^{(2)}\right) \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \left(\mathbf{F}^{(1)} + \widetilde{\mathbf{F}}^{(1)}\right) \cdot \frac{\partial}{\partial \mathbf{p}}\left(f_e^{(1)} + \widetilde{f}_e^{(1)}\right)$$

So using the linear propriety it becomes:

$$\mathcal{L}\check{f}_e^{(2)} + \mathcal{L} f_e^{(2)} + \mathcal{L}\widetilde{f}_e^{(2)} = -\check{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} -$$

$$-\mathbf{F}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}}$$

And then:

$$\mathcal{L}\check{f}_e^{(2)} + \mathcal{L} f_e^{(2)} + \mathcal{L}\widetilde{f}_e^{(2)} = -\check{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} -$$

$$-\left(\mathbf{F}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} + \mathbf{F}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}}\right) - \left(\widetilde{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} + \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}}\right) \tag{136}$$

Elements $\left[f_e^{(2)}, \mathbf{E}^{(2)}, \mathbf{B}^{(2)}\right]$ and $\left[\widetilde{f}_e^{(2)}, \widetilde{\mathbf{E}}^{(2)}, \widetilde{\mathbf{B}}^{(2)}\right]$ are singularly solution the second order Vlasov equation:

$$\mathcal{L} f_e^{(2)} = -\mathbf{F}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{137}$$

$$\mathcal{L}\widetilde{f}_e^{(2)} = -\widetilde{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} \tag{138}$$

So using the equations (137) and (138) for simplifying the (136) we obtain:

$$\mathcal{L}\check{f}_e^{(2)} = -\check{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{139}$$

that is the equation (43). We can represent $\check{f}_e^{(2)}$ as $\check{f}_e{}^{(2)} = \check{f}_{lin}^{(2)} + \check{f}_{int}^{(2)}$, remenbering the linearity of $\mathcal{L}$we have

$$\mathcal{L}\check{f}_e^{(2)} = \mathcal{L}\left(\check{f}_{lin}^{(2)} + \check{f}_{int}^{(2)}\right) = \mathcal{L}\check{f}_{lin}^{(2)} + \mathcal{L}\check{f}_{int}^{(2)} =$$

$$= -\check{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} - \mathbf{F}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{140}$$

$\check{f}_{lin}^{(2)}$ satisfies the part of the equation (141) linear in second order quantities:

$$\mathcal{L}\check{f}_{lin}^{(2)} = -\check{\mathbf{F}}^{(2)} \cdot \frac{\partial f_e^{(0)}}{\partial \mathbf{p}} \tag{141}$$

While $\check{f}_{int}^{(2)}$satisfies the bilinear part:

$$\mathcal{L}\check{f}_{int}^{(2)} = -\mathbf{F}^{(1)} \cdot \frac{\partial \widetilde{f}_e^{(1)}}{\partial \mathbf{p}} - \widetilde{\mathbf{F}}^{(1)} \cdot \frac{\partial f_e^{(1)}}{\partial \mathbf{p}} \tag{142}$$

So we have two currents:

$$\check{\mathbf{J}}_{lin}^{(1)} = q_e \int \mathbf{v}\check{f}_{lin}^{(2)} \mathrm{d}\mathbf{p} \qquad \check{\mathbf{J}}_{int}^{(2)} = q_e \int \mathbf{v}\check{f}_{int}^{(2)} \mathrm{d}\mathbf{p} \tag{143}$$

So $\check{\mathbf{J}}_{lin}^{(2)}$ is the second order current due to the linear response of the plasma to the field of the scattered wave, and $\mathbf{J}_{int}^{(2)}$ is the current due to the interaction of the incident wave with the fluctuations. The electromagnetic field of the scattered wave is then solution of the second-order wave equation (98):

$$\nabla \times \nabla \times \check{\mathbf{E}}^{(2)} + \varepsilon_0 \mu_0 \frac{\partial^2 \check{\mathbf{E}}^{(2)}}{\partial \mathrm{t}^2} = -\mu_0 \frac{\partial}{\partial \mathrm{t}} \left(\check{\mathbf{J}}_{lin}^{(2)} + \check{\mathbf{J}}_{int}^{(2)}\right) \tag{144}$$

Appendix D

We now derive in details the generalized fluid equations (53),(54) and (55) by the moments analysis of the Vlasov equation:

$$\frac{\partial f_e}{\partial t} + \mathbf{v} \cdot \frac{\partial f_e}{\partial \mathbf{r}} + q_e \left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right) \cdot \frac{\partial f_e}{\partial \mathbf{p}} = 0 \tag{145}$$

The moment zero of the Vlasov equation is then:

$$\int \frac{\partial f_e}{\partial t} \mathrm{d}\mathbf{p} + \int \mathbf{v} \cdot \nabla f_e \mathrm{d}\mathbf{p} + q_e \int \left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right) \cdot \frac{\partial f_e}{\partial \mathbf{p}} \mathrm{d}\mathbf{p} = 0 \tag{146}$$

We now analyze term by term this equation:

$$\int \frac{\partial f_e}{\partial t} \mathrm{d}\mathbf{p} = \frac{\partial}{\partial t} \int f_e \mathrm{d}\mathbf{p} = \frac{\partial n_e}{\partial t} \tag{147}$$

$$\int \mathbf{v} \cdot \nabla f_e \mathrm{d}\mathbf{p} = \nabla \cdot \int \mathbf{v} f_e \mathrm{d}\mathbf{p} = \nabla \cdot \overline{\mathbf{v}} \tag{148}$$

$$\int \mathbf{E} \cdot \frac{\partial f_e}{\partial \mathbf{p}} \mathrm{d}\mathbf{p} = \int \frac{\partial}{\partial \mathbf{p}} \cdot (\mathbf{E} f_e) \, \mathrm{d}\mathbf{p} = \int\limits_{S_\infty} \mathbf{E} f_e \mathrm{d}\mathbf{S} = 0 \tag{149}$$

$$\int (\mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f_e}{\partial \mathbf{p}} \mathrm{d}\mathbf{p} = \int \frac{\partial}{\partial \mathbf{p}} \left(f_e \mathbf{v} \times \mathbf{B} \right) \mathrm{d}\mathbf{p} - \int f_e \frac{\partial}{\partial \mathbf{p}} \left(\mathbf{v} \times \mathbf{B} \right) \mathrm{d}\mathbf{p} =$$

$$= \int\limits_{S_\infty} \left(f_e \mathbf{v} \times \mathbf{B} \right) \mathrm{d}\mathbf{S} = 0 \tag{150}$$

For the two surfaces integrals we have assumed the surface S_∞ to the infinity, the surface area increases as v^2, but a real distribution f_e vanishes much faster for $v \to \infty$ so the intergal on the infinity surfaces also vanishes. The second integral of the last term is zero because $\partial / \partial \mathbf{p}$ is perpendicular to $\mathbf{v} \times \mathbf{B}$. So the moment zero (146) is the continuity equation:

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \overline{\mathbf{v}} = 0 \tag{151}$$

The moment one of the Vlasov equation is:

$$\int \mathbf{v} \frac{\partial f_e}{\partial t} \mathrm{d}\mathbf{p} + \int \mathbf{v} \left(\mathbf{v} \cdot \nabla f_e \right) \mathrm{d}\mathbf{p} + q_e \int \mathbf{v} \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} \right) \cdot \frac{\partial f_e}{\partial \mathbf{p}} \mathrm{d}\mathbf{p} = 0 \tag{152}$$

Term by term:

$$\int \mathbf{v} \frac{\partial f_e}{\partial t} \mathrm{d}\mathbf{p} = \frac{\partial}{\partial t} \int \mathbf{v} f_e \mathrm{d}\mathbf{p} = \frac{\partial \overline{\mathbf{v}}}{\partial t} \tag{153}$$

$$\int \mathbf{v} \left(\mathbf{v} \cdot \nabla f_e \right) \mathrm{d}\mathbf{p} = \nabla \cdot \int \mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} = \nabla \cdot \overline{\mathbf{v}\mathbf{v}} \tag{154}$$

$$\int \mathbf{v} \left(\mathbf{E} \cdot \frac{\partial f_e}{\partial \mathbf{p}} \right) \mathrm{d}\mathbf{p} = \int \frac{\partial}{\partial \mathbf{p}} \cdot \left(f_e \mathbf{v} \mathbf{E} \right) \mathrm{d}\mathbf{p} - \frac{1}{m_e} \int f_e \mathbf{E} \mathrm{d}\mathbf{p} =$$

$$= \int\limits_{S_\infty} \left(f_e \mathbf{p} \mathbf{E} \right) \mathrm{d}\mathbf{S} - \frac{\mathbf{E}}{m_e} \int f_e \mathrm{d}\mathbf{p} = -\frac{n_e \mathbf{E}}{m_e} \tag{155}$$

$$\int \mathbf{v} \left(\mathbf{v} \times \mathbf{B} \right) \cdot \frac{\partial f_e}{\partial \mathbf{p}} \mathrm{d}\mathbf{p} = \int \frac{\partial}{\partial \mathbf{p}} \left[f_e \mathbf{v} \left(\mathbf{v} \times \mathbf{B} \right) \right] \mathrm{d}\mathbf{p} - \frac{1}{m_e} \int f_e \left(\mathbf{v} \times \mathbf{B} \right) \mathrm{d}\mathbf{p} -$$

$$-\int f_e \mathbf{v} \frac{\partial}{\partial \mathbf{p}} \cdot (\mathbf{v} \times \mathbf{B}) \, \mathrm{d}\mathbf{p} = \int\limits_{S_\infty} \left[f_e \mathbf{v} \left(\mathbf{v} \times \mathbf{B} \right) \right] \mathrm{d}\mathbf{S} - \frac{1}{m_e} \int \mathbf{v} f_e \mathrm{d}\mathbf{p} \times \mathbf{B} = -\frac{1}{m_e} \left(\overline{\mathbf{v}} \times \mathbf{B} \right) \tag{156}$$

So the moment one of the Vlasov equation is the equation for motion:

$$\frac{\partial \overline{\mathbf{v}}}{\partial t} + \nabla \cdot \overline{\mathbf{v}\mathbf{v}} - \frac{q_e}{m_e} \left(n_e \mathbf{E} + \overline{\mathbf{v}} \times \mathbf{B} \right) = 0 \tag{157}$$

The second moment is:

$$\int \mathbf{v}\mathbf{v} \frac{\partial f_e}{\partial t} \mathrm{d}\mathbf{p} + \int \mathbf{v}\mathbf{v} \left(\mathbf{v} \cdot \nabla f_e \right) \mathrm{d}\mathbf{p} + q_e \int \mathbf{v}\mathbf{v} \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} \right) \cdot \frac{\partial f_e}{\partial \mathbf{p}} \mathrm{d}\mathbf{p} = 0 \tag{158}$$

Term by term:

$$\int \mathbf{v}\mathbf{v} \frac{\partial f_e}{\partial t} \mathrm{d}\mathbf{p} = \frac{\partial}{\partial t} \int \mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} = \frac{\partial \overline{\mathbf{v}\mathbf{v}}}{\partial t} \tag{159}$$

$$\int \mathbf{v}\mathbf{v} \left(\mathbf{v} \cdot \nabla f_e \right) \mathrm{d}\mathbf{p} = \nabla \cdot \int \mathbf{v}\mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} = \nabla \cdot \overline{\mathbf{v}\mathbf{v}\mathbf{v}} \tag{160}$$

$$\int \mathbf{v}\mathbf{v} \left(\mathbf{E} \cdot \frac{\partial f_e}{\partial \mathbf{p}} \right) \mathrm{d}\mathbf{p} = \int \frac{\partial}{\partial \mathbf{p}} \cdot \left(f_e \mathbf{v}\mathbf{v}\mathbf{E} \right) \mathrm{d}\mathbf{p} - \frac{1}{m_e} \int f_e \mathbf{v}\mathbf{E} \mathrm{d}\mathbf{p} - \frac{1}{m_e} \int f_e \mathbf{E}\mathbf{v} \mathrm{d}\mathbf{p} =$$

$$= \int\limits_{S_\infty} \left(f_e \mathbf{v}\mathbf{v}\mathbf{E} \right) \mathrm{d}\mathbf{S} - \int \mathbf{v} f_e \mathrm{d}\mathbf{p} \frac{\mathbf{E}}{m_e} - \frac{\mathbf{E}}{m_e} \int \mathbf{v} f_e \mathrm{d}\mathbf{p} = -\frac{1}{m_e} \left(\overline{\mathbf{v}}\mathbf{E} + \mathbf{E}\overline{\mathbf{v}} \right) \tag{161}$$

$$\int \mathbf{v}\mathbf{v} \left(\mathbf{v} \times \mathbf{B} \right) \cdot \frac{\partial f_e}{\partial \mathbf{p}} \mathrm{d}\mathbf{p} = \int \frac{\partial}{\partial \mathbf{p}} \left[f_e \mathbf{v}\mathbf{v} \left(\mathbf{v} \times \mathbf{B} \right) \right] \mathrm{d}\mathbf{p} - \frac{1}{m_e} \int f_e \mathbf{v} \left(\mathbf{v} \times \mathbf{B} \right) \mathrm{d}\mathbf{p} -$$

$$-\frac{1}{m_e} \int f_e \left(\mathbf{v} \times \mathbf{B} \right) \mathbf{v} \mathrm{d}\mathbf{p} - \int f_e \mathbf{v}\mathbf{v} \frac{\partial}{\partial \mathbf{p}} \cdot (\mathbf{v} \times \mathbf{B}) = \int\limits_{S_\infty} \left[f_e \mathbf{v}\mathbf{v} \left(\mathbf{v} \times \mathbf{B} \right) \right] \mathrm{d}\mathbf{S} - \frac{1}{m_e} \int \mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} \times \mathbf{B} +$$

$$+\frac{1}{m_e} \mathbf{B} \times \int \mathbf{v}\mathbf{v} f_e \mathrm{d}\mathbf{p} = -\frac{1}{m_e} \left(\overline{\mathbf{v}\mathbf{v}} \times \mathbf{B} - \mathbf{B} \times \overline{\mathbf{v}\mathbf{v}} \right) \tag{162}$$

So the second moment of the Vlasov equation is the generalized energy transport equation:

$$\frac{\partial \overline{\mathbf{v}\mathbf{v}}}{\partial t} + \nabla \cdot \overline{\mathbf{v}\mathbf{v}\mathbf{v}} - \frac{q_e}{m_e} \left[\overline{\mathbf{v}}\mathbf{E} + \mathbf{E}\overline{\mathbf{v}} + \left(\overline{\mathbf{v}\mathbf{v}} \times \mathbf{B} \right) - \left(\mathbf{B} \times \overline{\mathbf{v}\mathbf{v}} \right) \right] = 0 \tag{163}$$

It is interesting to note that starting from this general equation it is possible to derive an explicit equation for the energy transport, infact the kinetic energy density $\varepsilon$is determined by the sum of the diagonal terms of the diad $\overline{\mathbf{vv}}$:

$$\varepsilon = \frac{m_e}{2}\int v^2 f_e \mathrm{d}\mathbf{p} = \frac{m_e}{2}\int \left(v_1^2 + v_2^2 + v_3^2\right) f_e \mathrm{d}\mathbf{p} \tag{164}$$

The detailed derivation of the transport equation for ε starting by the (163) is described in reference [13].

Appendix E

In this appendix we derive in details the relation (104) between the tensor $\mathbb{T}$ and the susceptibility tensor $\mathbb{X}$. $\mathbb{T}$ is the tensor that exprimes the linearized equation for motion in the Fourier space with the matricial representetion. As we have shown in our model, the linearized equation for motion is the first order equation for motion and the first order quantities are due to the linear waves and to the linear response of the plasma to these linear sollecitations. So we rewrite the linearized equation for motion and the first order current:

$$\frac{\partial \mathbf{u}^{(1)}}{\partial t} = \frac{q_e}{m_e}\left(\mathbf{E}^{(1)} + \mathbf{u}^{(1)} \times \mathbf{B}^{(0)}\right) \tag{165}$$

$$\mathbf{J}^{(1)} = q_e n_e^{(0)} \mathbf{u}^{(1)} \tag{166}$$

In the Fourier the equation for motion becomes:

$$-i\omega \mathbf{u}^{(1)} - \frac{q_e B^{(0)}}{m_e}\left(\mathbf{u}^{(1)} \times \frac{\mathbf{B}^{(0)}}{B^{(0)}}\right) = \frac{q_e}{m_e}\mathbf{E}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega)$$

And then:

$$\mathbb{T}(\mathbf{k},\omega)\cdot\mathbf{u}^{(1)} = \frac{q_e}{m_e}\mathbf{E}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega) \tag{167}$$

Where $\mathbb{T}(\mathbf{k},\omega)$ is the tensor:

$$\mathbb{T}(\mathbf{k},\omega) = -i\omega\mathbb{I} - \omega_{ce}\left(\underline{\underline{\varepsilon}}^{LC}\cdot\frac{\mathbf{B}^{(0)}}{B^{(0)}}\right) \tag{168}$$

Where $\omega_{pe} = q_e B^{(0)}/m_e$ .The fluid velocity in the Fourier space is the given by the expression:

$$\mathbf{u}^{(1)} = \frac{q_e}{m_e}\mathbb{T}^{-1}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega) \tag{169}$$

and the current becomes:

$$\mathbf{J}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega) = \frac{q_e^2 n_e^{(0)}}{m_e}\mathbb{T}^{-1}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega) \tag{170}$$

Remembering that $\mathbf{J}_{\mathbf{k}}^{(1)}$is the current due to the linear propagatin of the wave in the medium, from the linear response theory we have:

$$\mathbf{J}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega)=\mathbb{S}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega)=-i\varepsilon_0\omega\mathbb{X}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega) \tag{171}$$

Equaling the (170) and the (171) we obtain:

$$\mathbb{T}^{-1}(\mathbf{k},\omega)=-i\frac{\varepsilon_0 m_e\omega}{q_e^2 n_e^{(0)}}\mathbb{X}(\mathbf{k},\omega)=-i\frac{\omega}{\omega_{pe}^2}\mathbb{X}(\mathbf{k},\omega) \tag{172}$$

Using this results in the (167) we can exprime the first order fluid velocity in function of the susceptibility tensor:

$$\mathbf{u}^{(1)}=-i\frac{\omega\varepsilon_0}{n_e^{(0)}q_e}\mathbb{X}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}^{(1)}(\mathbf{k},\omega) \tag{173}$$

So we have also demonstrated the relation (111).

Appendix F

We now calculate explicitely the mode coupling coefficient for the density fluctuations $V^n\left(\mathbf{k},\check{\mathbf{k}},\omega_k\right)$:

$$V^n\left(\mathbf{k},\check{\mathbf{k}},\omega_k\right)=\frac{\omega_k}{2}\frac{\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}(\mathbf{k})\right)\cdot\mathbf{e}(\mathbf{k})}{\left(\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\cdot\left(\mathbf{e}^\star(\mathbf{k})\cdot\mathbb{M}\cdot\mathbf{e}(\mathbf{k})\right)^{\frac{1}{2}}} \tag{174}$$

Remembering that $\mathbb{K}=\mathbb{I}+\mathbb{X}$, we can exprime this coefficint in function of the susceptibility tensor:

$$V^n\left(\mathbf{k},\check{\mathbf{k}},\omega\right)=-\frac{\omega_k}{2}\frac{\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot\mathbb{X}(\mathbf{k})\cdot\mathbf{e}(\mathbf{k})}{\left(\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\cdot\left(\mathbf{e}^\star(\mathbf{k})\cdot\mathbb{M}\cdot\mathbf{e}(\mathbf{k})\right)^{\frac{1}{2}}} \tag{175}$$

The numerator of the fraction is given by the product:

$$\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot\mathbb{X}(\mathbf{k})\cdot\mathbf{e}(\mathbf{k})=$$

$$=\check{e}_x^\star e_x X_{xx}+\check{e}_y^\star e_x X_{yx}+\check{e}_z^\star e_x X_{zx}+\check{e}_x^\star e_y X_{xy}+\check{e}_y^\star e_y X_{yy}+\check{e}_z^\star e_y X_{zy}+$$

$$+\check{e}_x^\star e_z X_{xz}+\check{e}_y^\star e_z X_{yz}+\check{e}_z^\star e_z X_{zz}=$$

$$=\check{e}_x^\star\left[e_x X_{xx}+e_y X_{xy}+e_z X_{xz}\right]+\check{e}_y^\star\left[e_x X_{yx}+e_y X_{yy}+e_z X_{yz}\right]+$$

$$+\check{e}_z^\star \left[e_x X_{zx} + e_y X_{zy} + e_z X_{zz}\right] \tag{176}$$

The two factors of the denominator are:

$$\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right) \cdot \mathbb{M} \cdot \check{\mathbf{e}}\left(\check{\mathbf{k}}\right) =$$

$$= \check{e}_x^\star \check{e}_x M_{xx} + \check{e}_y^\star \check{e}_x M_{yx} + \check{e}_z^\star \check{e}_x M_{zx} + \check{e}_x^\star \check{e}_y M_{xy} + \check{e}_y^\star \check{e}_y M_{yy} + \check{e}_z^\star \check{e}_y M_{zx} +$$

$$+ \check{e}_x^\star \check{e}_z M_{xz} + \check{e}_y^\star \check{e}_z M_{yz} + \check{e}_z^\star \check{e}_z M_{zz} =$$

$$= \check{e}_x^\star \left[\check{e}_x M_{xx} + \check{e}_y M_{xy} + \check{e}_z M_{xz}\right] + \check{e}_y^\star \left[\check{e}_x M_{yx} + \check{e}_y M_{yy} + \check{e}_z M_{yz}\right] +$$

$$+ \check{e}_z^\star \left[\check{e}_x M_{zx} + \check{e}_y M_{zy} + \check{e}_z M_{zz}\right] \tag{177}$$

$$\mathbf{e}^\star\left(\mathbf{k}\right) \cdot \mathbb{M} \cdot \mathbf{e}\left(\mathbf{k}\right) =$$

$$= e_x^\star e_x M_{xx} + e_y^\star e_x M_{yx} + e_z^\star e_x M_{zx} + e_x^\star e_y M_{xy} + e_y^\star e_y M_{yy} + e_z^\star e_y M_{zy} +$$

$$+ e_x^\star e_z M_{xz} + e_y^\star e_z M_{yz} + e_z^\star e_z M_{zz} =$$

$$= e_x^\star \left[e_x M_{xx} + e_y M_{xy} + e_z M_{xz}\right] + e_y^\star \left[e_x M_{yx} + e_y M_{yy} + e_z M_{yz}\right] +$$

$$+ e_z^\star \left[e_x M_{zx} + e_y M_{zy} + e_z M_{zz}\right] \tag{178}$$

We need an expression for the tensor $\mathbb{M}$ and for the polarization vector. Remembering that the dispersion tensor is given by:

$$\mathbb{D} = \begin{pmatrix} K_{xx} - N_\parallel^2 & K_{xy} & K_{xz} + N_\perp N_\parallel \\ -K_{xy} & K_{yy} - N^2 & K_{yz} \\ K_{xz} + N_\perp N_\parallel & -K_{yz} & K_{zz} - N_\perp^2 \end{pmatrix} \tag{179}$$

And that $\mathbb{K} = \mathbb{I} + \mathbb{X}$, the elements of the tensor $\mathbb{M}$ are:

$$
\begin{aligned}
M_{xx} &= \frac{\omega_k}{2}\frac{\partial D_{xx}}{\partial \omega} = \frac{\omega_k}{2}\frac{\partial X_{xx}}{\partial \omega} + N_{\parallel}^2 \\
M_{xy} &= \frac{\omega_k}{2}\frac{\partial D_{xy}}{\partial \omega} = \frac{\omega_k}{2}\frac{\partial X_{xy}}{\partial \omega} \\
M_{xz} &= \frac{\omega_k}{2}\frac{\partial D_{xz}}{\partial \omega} = \frac{\omega_k}{2}\frac{\partial X_{xz}}{\partial \omega} - N_{\perp}N_{\parallel} \\
M_{yx} &= \frac{\omega_k}{2}\frac{\partial D_{yx}}{\partial \omega} = -\frac{\omega_k}{2}\frac{\partial X_{xy}}{\partial \omega} \\
M_{yy} &= \frac{\omega_k}{2}\frac{\partial D_{yy}}{\partial \omega} = \frac{\omega_k}{2}\frac{\partial X_{yy}}{\partial \omega} + N^2 \\
M_{yz} &= \frac{\omega_k}{2}\frac{\partial D_{yz}}{\partial \omega} = \frac{\omega_k}{2}\frac{\partial X_{yz}}{\partial \omega} \\
M_{zx} &= \frac{\omega_k}{2}\frac{\partial D_{zx}}{\partial \omega} = \frac{\omega_k}{2}\frac{\partial X_{zx}}{\partial \omega} - N_{\perp}N_{\parallel} \\
M_{zy} &= \frac{\omega_k}{2}\frac{\partial D_{zy}}{\partial \omega} = -\frac{\omega_k}{2}\frac{\partial X_{yz}}{\partial \omega} \\
M_{zz} &= \frac{\omega_k}{2}\frac{\partial D_{zz}}{\partial \omega} = \frac{\omega_k}{2}\frac{\partial X_{zz}}{\partial \omega} + N_{\perp}^2
\end{aligned}
$$

So the tensor $\mathbb{M}$ is:

$$
\mathbb{M} = \frac{\omega_k}{2}\frac{\partial \mathbb{D}}{\partial \omega} = \begin{pmatrix}
\frac{\omega_k}{2}\frac{\partial X_{xx}}{\partial \omega} + N_{\parallel}^2 & \frac{\omega_k}{2}\frac{\partial X_{xy}}{\partial \omega} & \frac{\omega_k}{2}\frac{\partial X_{xz}}{\partial \omega} - N_{\perp}N_{\parallel} \\
-\frac{\omega_k}{2}\frac{\partial X_{xy}}{\partial \omega} & \frac{\omega_k}{2}\frac{\partial X_{yy}}{\partial \omega} + N^2 & \frac{\omega_k}{2}\frac{\partial X_{yz}}{\partial \omega} \\
\frac{\omega_k}{2}\frac{\partial X_{xz}}{\partial \omega} - N_{\perp}N_{\parallel} & -\frac{\omega_k}{2}\frac{\partial X_{yz}}{\partial \omega} & \frac{\omega_k}{2}\frac{\partial X_{zz}}{\partial \omega} + N_{\perp}^2
\end{pmatrix} \tag{180}
$$

The derivatives of the elements of the susceptibility tensor $\mathbb{X}\left(\mathbf{N}, \beta_{Ts}, \overline{\omega}_{ps}, \overline{\omega}_{cs}\right)$ respect to the frequency ω are given by the expressions:

$$
\begin{aligned}
\frac{\partial X_{xx}}{\partial \omega} &= \frac{\partial X_{xx}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial X_{xx}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial X_{xx}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial X_{xx}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial X_{xy}}{\partial \omega} &= \frac{\partial X_{xy}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial X_{xy}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial X_{xy}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial X_{xy}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial X_{xz}}{\partial \omega} &= \frac{\partial X_{xz}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial X_{xz}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial X_{xz}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial X_{xz}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial X_{yy}}{\partial \omega} &= \frac{\partial X_{yy}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial X_{yy}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial X_{yy}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial X_{yy}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial X_{yz}}{\partial \omega} &= \frac{\partial X_{yz}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial X_{yz}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial X_{yz}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial X_{yz}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial X_{zz}}{\partial \omega} &= \frac{\partial X_{zz}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial X_{zz}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial X_{zz}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial X_{zz}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega}
\end{aligned}
$$

The kernel of the scattering probability is given by the expression:

$$K\left(k_{\perp}^{\sigma},\check{k}_{\perp}^{\sigma},\beta\right)=\frac{2\pi\check{k}_{\perp}^{\sigma}}{v_{G\perp}^{\sigma}}\left|V^{n}\left(\mathbf{k}^{\sigma},\check{\mathbf{k}}^{\sigma},\beta\right)\right|^{2} \tag{181}$$

Where the group velocity is evaluated for the scattered wave vector. The group velocity for the mode σ is given by the expression:

$$\mathbf{v}_{G}^{\sigma}=\frac{d\mathbf{X}}{dt}=-\left.\frac{\partial\mathcal{D}/\partial\mathbf{k}}{\partial\mathcal{D}/\partial\omega}\right|_{\mathbf{k}=\mathbf{k}^{\sigma}} \tag{182}$$

We have demonstrated that the component of the group velocity perpendicular to the magnetic field can be calculated using the expression:

$$v_{G\perp}=-\left.\frac{\partial\mathcal{D}/\partial k_{\perp}}{\partial\mathcal{D}/\partial\omega}\right|_{k_{\perp}=k_{\perp}^{\sigma}} \tag{183}$$

Since $k_{\parallel}\simeq k_{\parallel}^{\sigma}$. So the kernel of the scattering probability becomes:

$$K\left(k_{\perp}^{\sigma},\check{k}_{\perp}^{\sigma},\beta\right)=-2\pi\check{k}_{\perp}^{\sigma}\left.\frac{\partial\mathcal{D}/\partial\omega}{\partial\mathcal{D}/\partial k_{\perp}}\right|_{k_{\perp}=k_{\perp}^{\sigma}}\cdot\left|V^{n}\left(\mathbf{k}^{\sigma},\check{\mathbf{k}}^{\sigma},\beta\right)\right| \tag{184}$$

Appendix G

We now calculate explicitely the mode coupling coefficient for the density fluctuations $\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right)$, it is important to note that while for density fluctuation the coupling coefficient is a scalar, for magnetic field's fluctuations it is a vector. We start from the expression, that is the term of the mode coupling equation due to the fluctuations of the magnetic field:

$$i\frac{\omega_{ce}\check{\omega}_{\check{k}}^{2}}{2\omega_{pe}^{2}}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\underline{\underline{\varepsilon}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}=\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right)\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}} \tag{185}$$

Remembering that $\mathbb{K}=\mathbb{I}+\mathbb{X}$, we can write:

$$i\frac{\omega_{ce}\check{\omega}_{\check{k}}^{2}}{2\omega_{pe}^{2}}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{X}\left(\check{\mathbf{k}}\right)\cdot\left(\underline{\underline{\varepsilon}}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}\right)\cdot\mathbb{X}\left(\mathbf{k}\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}=\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right)\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}} \tag{186}$$

The denominator of the (187) is a scalar and it has been calculated in the precedent appendix, so it is useful to write this equation in the form:

$$i\frac{\omega_{ce}\check{\omega}_{\check{k}}^2}{2\omega_{pe}^2}\frac{\mathbf{w}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right)\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}}}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}=\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right)\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B^{(0)}} \tag{187}$$

So the mode coupling coefficient is given by the expression:

$$i\frac{\omega_{ce}\check{\omega}_{\check{k}}^2}{2\omega_{pe}^2}\frac{\mathbf{w}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}=\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right) \tag{188}$$

We have just calculated the denominator, so we need an explicit expression for $\mathbf{W}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega\right)$. From this point we use the compact notation $\mathbb{X}\left(\mathbf{k}\right)=\mathbb{X},\mathbb{X}\left(\check{\mathbf{k}}\right)=\check{\mathbb{X}}$ and $\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)/B^{(0)}=\delta\mathbf{B}/B^{(0)}$. The vector $\mathbf{W}^{\mathbf{B}}$can be derived from the relation:

$$\check{\mathbf{e}}^{\star}\cdot\left[\check{\mathbb{X}}\cdot\left(\underset{\equiv}{\varepsilon}^{LC}\cdot\frac{\delta\mathbf{B}}{B^{(0)}}\right)\cdot\mathbb{X}\right]\cdot\mathbf{e}=\mathbf{w}^{\mathbf{B}}\cdot\frac{\delta\mathbf{B}}{B^{(0)}} \tag{189}$$

We now calculate explicitely this quantity and after we derive the components of the vector $\mathbf{w}^{\mathbf{B}}$. $\underset{\equiv}{\varepsilon}^{LC}$ is the Levi-Civita tensor of order 3, the scalar product of this tensor with the vector $\delta\mathbf{B}/B^{(0)}$gives a tensor of order 2:

$$\mathbb{U} = \underset{\equiv}{\varepsilon}^{LC}\cdot\frac{\delta\mathbf{B}}{B^{(0)}} \tag{190}$$

$$U_{ij} = \varepsilon_{ijk}\frac{\delta B_k}{B^{(0)}} \tag{191}$$

So the elements of $\mathbb{U}$ are:

$$\begin{aligned}
U_{xx} &= \varepsilon_{xxx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{xxy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{xxz}\frac{\delta B_z}{B^{(0)}}=0\\
U_{xy} &= \varepsilon_{xyx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{xyy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{xyz}\frac{\delta B_z}{B^{(0)}}=\frac{\delta B_z}{B^{(0)}}\\
U_{xz} &= \varepsilon_{xzx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{xzy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{xzz}\frac{\delta B_z}{B^{(0)}}=-\frac{\delta B_y}{B^{(0)}}\\
U_{yx} &= \varepsilon_{yxx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{yxy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{yxz}\frac{\delta B_z}{B^{(0)}}=-\frac{\delta B_z}{B^{(0)}}\\
U_{yy} &= \varepsilon_{yyx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{yyy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{yyz}\frac{\delta B_z}{B^{(0)}}=0\\
U_{yz} &= \varepsilon_{yzx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{yzy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{yzz}\frac{\delta B_z}{B^{(0)}}=\frac{\delta B_x}{B^{(0)}}\\
U_{zx} &= \varepsilon_{zxx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{zxy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{zxz}\frac{\delta B_z}{B^{(0)}}=\frac{\delta B_z}{B^{(0)}}\\
U_{zy} &= \varepsilon_{zyx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{zyy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{zyz}\frac{\delta B_z}{B^{(0)}}=-\frac{\delta B_x}{B^{(0)}}\\
U_{zz} &= \varepsilon_{zzx}\frac{\delta B_x}{B^{(0)}}+\varepsilon_{zzy}\frac{\delta B_y}{B^{(0)}}+\varepsilon_{zzz}\frac{\delta B_z}{B^{(0)}}=0
\end{aligned}$$

So the tensor $\mathbb{U}$ is:

$$\mathbb{U} = \left(\underset{\equiv}{\epsilon}^{LC} \cdot \frac{\delta \mathbf{B}}{B^{(0)}} \right) = \begin{pmatrix} 0 & \frac{\delta B_z}{B^{(0)}} & -\frac{\delta B_y}{B^{(0)}} \\ -\frac{\delta B_z}{B^{(0)}} & 0 & \frac{\delta B_x}{B^{(0)}} \\ \frac{\delta B_y}{B^{(0)}} & -\frac{\delta B_x}{B^{(0)}} & 0 \end{pmatrix} \tag{192}$$

We now have to calculate the tensor $\mathbb{W}$, given by the expression:

$$\mathbb{W} = \check{\mathbb{X}} \cdot \mathbb{U} \cdot \mathbb{X} \tag{193}$$

The elements of $\mathbb{W}$ are:

$$\begin{aligned}
W_{xx} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{xy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{xz} \right) X_{xx} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{xx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{xz} \right) X_{yx} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{xx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{xy} \right) X_{zx} \\
W_{xy} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{xy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{xz} \right) X_{xy} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{xx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{xz} \right) X_{yy} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{xx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{xy} \right) X_{zy} \\
W_{xz} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{xy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{xz} \right) X_{xz} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{xx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{xz} \right) X_{yz} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{xx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{xy} \right) X_{zz} \\
W_{yx} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{yy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{yz} \right) X_{xx} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{yx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{yz} \right) X_{yx} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{yx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{yy} \right) X_{zx} \\
W_{yy} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{yy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{yz} \right) X_{xy} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{yx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{yz} \right) X_{yy} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{yx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{yy} \right) X_{zy} \\
W_{yz} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{yy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{yz} \right) X_{xz} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{yx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{yz} \right) X_{yz} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{yx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{yy} \right) X_{zz} \\
W_{zx} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{zy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{zz} \right) X_{xx} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{zx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{zz} \right) X_{yx} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{zx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{zy} \right) X_{zx} \\
W_{zy} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{zy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{zz} \right) X_{xy} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{zx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{zz} \right) X_{yy} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{zx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{zy} \right) X_{zy} \\
W_{zz} &= \left(-\frac{\delta B_z}{B^{(0)}} \check{X}_{zy} + \frac{\delta B_y}{B^{(0)}} \check{X}_{zz} \right) X_{xz} + \left(\frac{\delta B_z}{B^{(0)}} \check{X}_{zx} - \frac{\delta B_x}{B^{(0)}} \check{X}_{zz} \right) X_{yz} + \left(-\frac{\delta B_y}{B^{(0)}} \check{X}_{zx} + \frac{\delta B_x}{B^{(0)}} \check{X}_{zy} \right) X_{zz}
\end{aligned}$$

It is useful to write the elements of $\mathbb{W}$ in the form:

$$
\begin{aligned}
W_{xx} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{xy}X_{zx}-\check{X}_{xz}X_{yx}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{xz}X_{xx}-\check{X}_{xx}X_{zx}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{xx}X_{yx}-\check{X}_{xy}X_{xx}\right)\\
W_{xy} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{xy}X_{zy}-\check{X}_{xz}X_{yy}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{xz}X_{xy}-\check{X}_{xx}X_{zy}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{xx}X_{yy}-\check{X}_{xy}X_{xy}\right)\\
W_{xz} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{xy}X_{zz}-\check{X}_{xz}X_{yz}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{xz}X_{xz}-\check{X}_{xx}X_{zz}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{xx}X_{yz}-\check{X}_{xy}X_{xz}\right)\\
W_{yx} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{yy}X_{zx}-\check{X}_{yz}X_{yx}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{yz}X_{xx}-\check{X}_{yx}X_{zx}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{yx}X_{yx}-\check{X}_{yy}X_{xx}\right)\\
W_{yy} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{yy}X_{zy}-\check{X}_{yz}X_{yy}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{yz}X_{xy}-\check{X}_{yx}X_{zy}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{yx}X_{yy}-\check{X}_{yy}X_{xy}\right)\\
W_{yz} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{yy}X_{zz}-\check{X}_{yz}X_{yz}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{yz}X_{xz}-\check{X}_{yx}X_{zz}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{yx}X_{yz}-\check{X}_{yy}X_{xz}\right)\\
W_{zx} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{zy}X_{zx}-\check{X}_{zz}X_{yx}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{zz}X_{xx}-\check{X}_{zx}X_{zx}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{zx}X_{yx}-\check{X}_{zy}X_{xx}\right)\\
W_{zy} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{zy}X_{zy}-\check{X}_{zz}X_{yy}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{zz}X_{xy}-\check{X}_{zx}X_{zy}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{zx}X_{yy}-\check{X}_{zy}X_{xy}\right)\\
W_{zz} &= \frac{\delta B_x}{B^{(0)}}\left(\check{X}_{zy}X_{zz}-\check{X}_{zz}X_{yz}\right)+\frac{\delta B_y}{B^{(0)}}\left(\check{X}_{zz}X_{xz}-\check{X}_{zx}X_{zz}\right)+\frac{\delta B_z}{B^{(0)}}\left(\check{X}_{zx}X_{yz}-\check{X}_{zy}X_{xz}\right)
\end{aligned}
$$

Using these components of $\mathbb{W}$ we now calculate:

$$
\mathbf{w}^{\mathbf{B}}\cdot\frac{\delta\mathbf{B}}{B^{(0)}}=\check{\mathbf{e}}^{\star}\cdot\left[\check{\mathbb{X}}\cdot\left(\underset{\equiv}{\varepsilon}^{LC}\cdot\frac{\delta\mathbf{B}}{B^{(0)}}\right)\cdot\mathbb{X}\right]\cdot\mathbf{e}=\check{\mathbf{e}}^{\star}\cdot\mathbb{W}\cdot\mathbf{e}=
$$

$$
=\check{e}_x^{\star}e_xW_{xx}+\check{e}_y^{\star}e_xW_{yx}+\check{e}_z^{\star}e_xW_{zx}+\check{e}_x^{\star}e_yW_{xy}+\check{e}_y^{\star}e_yW_{yy}+\check{e}_z^{\star}e_yW_{zy}+
$$

$$
+\check{e}_x^{\star}e_zW_{xz}+\check{e}_y^{\star}e_zW_{yz}+\check{e}_z^{\star}e_zW_{zz}=
$$

$$
\begin{aligned}
= &\ \check{e}_x^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{xy}X_{zx}-\check{X}_{xz}X_{yx}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{xz}X_{xx}-\check{X}_{xx}X_{zx}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{xx}X_{yx}-\check{X}_{xy}X_{xx}\right)\right]e_x &+\\
+ &\ \check{e}_y^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{yy}X_{zx}-\check{X}_{yz}X_{yx}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{yz}X_{xx}-\check{X}_{yx}X_{zx}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{yx}X_{yx}-\check{X}_{yy}X_{xx}\right)\right]e_x &+\\
+ &\ \check{e}_z^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{zy}X_{zx}-\check{X}_{zz}X_{yx}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{zz}X_{xx}-\check{X}_{zx}X_{zx}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{zx}X_{yy}-\check{X}_{zy}X_{xx}\right)\right]e_x &+\\
+ &\ \check{e}_x^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{xy}X_{zy}-\check{X}_{xz}X_{yy}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{xz}X_{xy}-\check{X}_{xx}X_{zy}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{xx}X_{yy}-\check{X}_{xy}X_{xy}\right)\right]e_y &+\\
+ &\ \check{e}_y^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{yy}X_{zy}-\check{X}_{yz}X_{yy}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{yz}X_{xy}-\check{X}_{yx}X_{zy}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{yx}X_{yy}-\check{X}_{yy}X_{xy}\right)\right]e_y &+\\
+ &\ \check{e}_z^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{zy}X_{zy}-\check{X}_{zz}X_{yy}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{zz}X_{xy}-\check{X}_{zx}X_{zy}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{zx}X_{yy}-\check{X}_{zy}X_{xy}\right)\right]e_y &+\\
+ &\ \check{e}_x^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{xy}X_{zz}-\check{X}_{xz}X_{yz}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{xz}X_{xz}-\check{X}_{xx}X_{zz}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{xx}X_{yz}-\check{X}_{xy}X_{xz}\right)\right]e_z &+\\
+ &\ \check{e}_y^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{yy}X_{zz}-\check{X}_{yz}X_{yz}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{yz}X_{xz}-\check{X}_{yx}X_{zz}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{yx}X_{yz}-\check{X}_{yy}X_{xz}\right)\right]e_z &+\\
+ &\ \check{e}_z^{\star}\left[\tfrac{\delta B_x}{B^{(0)}}\left(\check{X}_{zy}X_{zz}-\check{X}_{zz}X_{yz}\right)+\tfrac{\delta B_y}{B^{(0)}}\left(\check{X}_{zz}X_{xz}-\check{X}_{zx}X_{zz}\right)+\tfrac{\delta B_z}{B^{(0)}}\left(\check{X}_{zx}X_{yz}-\check{X}_{zy}X_{xz}\right)\right]e_z &=
\end{aligned}
$$

$$= \frac{\delta B_x}{B^{(0)}} \left[\check{e}_x^\star e_x \left(\check{X}_{xy} X_{zx} - \check{X}_{xz} X_{yx} \right) + \check{e}_y^\star e_x \left(\check{X}_{yy} X_{zx} - \check{X}_{yz} X_{yx} \right) + \check{e}_z^\star e_x \left(\check{X}_{zy} X_{zx} - \check{X}_{zz} X_{yx} \right) + \right.$$

$$+ \check{e}_x^\star e_y \left(\check{X}_{xy} X_{zy} - \check{X}_{xz} X_{yy} \right) + \check{e}_y^\star e_y \left(\check{X}_{yy} X_{zy} - \check{X}_{yz} X_{yy} \right) + \check{e}_z^\star e_y \left(\check{X}_{zy} X_{zy} - \check{X}_{zz} X_{yy} \right) +$$

$$\left. + \check{e}_x^\star e_z \left(\check{X}_{xy} X_{zz} - \check{X}_{xz} X_{yz} \right) + \check{e}_y^\star e_z \left(\check{X}_{yy} X_{zz} - \check{X}_{yz} X_{yz} \right) + \check{e}_z^\star e_z \left(\check{X}_{xy} X_{zz} - \check{X}_{zz} X_{yz} \right) \right] +$$

$$+ \frac{\delta B_y}{B^{(0)}} \left[\check{e}_x^\star e_x \left(\check{X}_{xz} X_{xx} - \check{X}_{xx} X_{zx} \right) + \check{e}_y^\star e_x \left(\check{X}_{yz} X_{xx} - \check{X}_{yx} X_{zx} \right) + \check{e}_z^\star e_x \left(\check{X}_{zz} X_{xx} - \check{X}_{zx} X_{zx} \right) + \right.$$

$$+ \check{e}_x^\star e_y \left(\check{X}_{xz} X_{xy} - \check{X}_{xx} X_{zy} \right) + \check{e}_y^\star e_y \left(\check{X}_{yz} X_{xx} + - \check{X}_{yx} X_{zy} \right) + \check{e}_z^\star e_y \left(\check{X}_{zz} X_{xy} - \check{X}_{zx} X_{zy} \right) +$$

$$\left. + \check{e}_x^\star e_z \left(\check{X}_{xz} X_{xz} - \check{X}_{xx} X_{zz} \right) + \check{e}_y^\star e_z \left(\check{X}_{yz} X_{xz} - \check{X}_{yx} X_{zz} \right) + \check{e}_z^\star e_z \left(\check{X}_{zz} X_{xz} - \check{X}_{zx} X_{zz} \right) \right] +$$

$$+ \frac{\delta B_z}{B^{(0)}} \left[\check{e}_x^\star e_x \left(\check{X}_{xx} X_{yx} - \check{X}_{xy} X_{xx} \right) + \check{e}_y^\star e_x \left(\check{X}_{yx} X_{yx} - \check{X}_{yy} X_{xx} \right) + \check{e}_z^\star e_x \left(\check{X}_{zx} X_{yx} - \check{X}_{zy} X_{xx} \right) + \right.$$

$$+ \check{e}_x^\star e_y \left(\check{X}_{xx} X_{yy} - \check{X}_{xy} X_{xy} \right) + \check{e}_y^\star e_y \left(\check{X}_{yx} X_{yy} - \check{X}_{yy} X_{xy} \right) + \check{e}_z^\star e_y \left(\check{X}_{zx} X_{yy} - \check{X}_{zy} X_{xy} \right) +$$

$$\left. + \check{e}_x^\star e_z \left(\check{X}_{xx} X_{yz} - \check{X}_{xy} X_{xz} \right) + \check{e}_y^\star e_z \left(\check{X}_{yx} X_{yz} - \check{X}_{yy} X_{xz} \right) + \check{e}_z^\star e_z \left(\check{X}_{zx} X_{yz} - \check{X}_{zy} X_{yxz} \right) \right] =$$

$$= \frac{\delta B_x}{B^{(0)}} w_x^{\mathbf{B}} + \frac{\delta B_y}{B^{(0)}} w_y^{\mathbf{B}} + \frac{\delta B_z}{B^{(0)}} w_z^{\mathbf{B}} \tag{194}$$

So we now have the components of the vector $\mathbf{w}^{\mathbf{B}}$:

$$w_x^{\mathbf{B}} = \check{e}_x^\star e_x \left(\check{X}_{xy} X_{zx} - \check{X}_{xz} X_{yx} \right) + \check{e}_y^\star e_x \left(\check{X}_{yy} X_{zx} - \check{X}_{yz} X_{yx} \right) + \check{e}_z^\star e_x \left(\check{X}_{zy} X_{zx} - \check{X}_{zz} X_{yx} \right) +$$

$$+ \check{e}_x^\star e_y \left(\check{X}_{xy} X_{zy} - \check{X}_{xz} X_{yy} \right) + \check{e}_y^\star e_y \left(\check{X}_{yy} X_{zy} - \check{X}_{yz} X_{yy} \right) + \check{e}_z^\star e_y \left(\check{X}_{zy} X_{zy} - \check{X}_{zz} X_{yy} \right) +$$

$$+ \check{e}_x^\star e_z \left(\check{X}_{xy} X_{zz} - \check{X}_{xz} X_{yz} \right) + \check{e}_y^\star e_z \left(\check{X}_{yy} X_{zz} - \check{X}_{yz} X_{yz} \right) + \check{e}_z^\star e_z \left(\check{X}_{xy} X_{zz} - \check{X}_{zz} X_{yz} \right) =$$

$$= e_x X_{zx}\left(\check{e}_x^\star \check{X}_{xy} + \check{e}_y^\star \check{X}_{yy} + \check{e}_z^\star \check{X}_{zy}\right) - e_x X_{yx}\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y^\star \check{X}_{yz} + \check{e}_z^\star \check{X}_{zz}\right) +$$

$$+ e_y X_{zy}\left(\check{e}_x^\star \check{X}_{xy} + \check{e}_y^\star \check{X}_{yy} + \check{e}_z^\star \check{X}_{zy}\right) - e_y X_{yy}\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y^\star \check{X}_{yz} + \check{e}_z^\star \check{X}_{zz}\right) +$$

$$+ e_z X_{zz}\left(\check{e}_x^\star \check{X}_{xy} + \check{e}_y^\star \check{X}_{yy} + \check{e}_z^\star X_{zy}\right) - e_y X_{yz}\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y^\star \check{X}_{yz} + \check{e}_z^\star \check{X}_{zz}\right) =$$

$$= \left(e_x X_{zx} + e_y X_{zy} + e_z X_{zz}\right)\left(\check{e}_x^\star \check{X}_{xy} + \check{e}_y \check{X}_{yy} + \check{e}_z^\star \check{X}_{zy}\right) -$$

$$- \left(e_x X_{yx} + e_y X_{yy} + e_z X_{yz}\right)\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y^\star \check{X}_{yz} + \check{e}_z^\star \check{X}_{zz}\right) \qquad (195)$$

$$w_y^{\mathbf{B}} = \check{e}_x^\star e_x\left(\check{X}_{xz}X_{xx} - \check{X}_{xx}X_{zx}\right) + \check{e}_y^\star e_x\left(\check{X}_{yz}X_{xx} - \check{X}_{yx}X_{zx}\right) + \check{e}_z^\star e_x\left(\check{X}_{zz}X_{xx} - \check{X}_{zx}X_{zx}\right) +$$

$$+ \check{e}_x^\star e_y\left(\check{X}_{xz}X_{xy} - \check{X}_{xx}X_{zy}\right) + \check{e}_y^\star e_y\left(\check{X}_{yz}X_{xx} - \check{X}_{yx}X_{zy}\right) + \check{e}_z^\star e_y\left(\check{X}_{zz}X_{xy} - \check{X}_{zx}X_{zy}\right) +$$

$$+ \check{e}_x^\star e_z\left(\check{X}_{xz}X_{xz} - \check{X}_{xx}X_{zz}\right) + \check{e}_y^\star e_z\left(\check{X}_{yz}X_{xz} - \check{X}_{yx}X_{zz}\right) + \check{e}_z^\star e_z\left(\check{X}_{zz}X_{xz} - \check{X}_{zx}X_{zz}\right) =$$

$$= e_x X_{xx}\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y^\star \check{X}_{yz} + \check{e}_z^\star \check{X}_{zz}\right) - e_x X_{zx}\left(\check{e}_x^\star \check{X}_{xx} + \check{e}_y^\star \check{X}_{yx} + \check{e}_z^\star \check{X}_{zx}\right) +$$

$$+ e_y X_{xy}\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y^\star \check{X}_{yz} + \check{e}_z^\star \check{X}_{zz}\right) - e_y X_{zy}\left(\check{e}_x^\star \check{X}_{xx} + \check{e}_y^\star \check{X}_{yx} + \check{e}_z^\star \check{X}_{zx}\right) +$$

$$+ e_z X_{xz}\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y^\star \check{X}_{yz} + \check{e}_z^\star X_{zz}\right) - e_y X_{zz}\left(\check{e}_x^\star \check{X}_{xx} + \check{e}_y^\star \check{X}_{yx} + \check{e}_z^\star \check{X}_{zx}\right) =$$

$$= \left(e_x X_{xx} + e_y X_{xy} + e_z X_{xz}\right)\left(\check{e}_x^\star \check{X}_{xz} + \check{e}_y \check{X}_{yz} + \check{e}_z^\star \check{X}_{zz}\right) -$$

$$- \left(e_x X_{zx} + e_y X_{zy} + e_z X_{zz}\right)\left(\check{e}_x^\star \check{X}_{xx} + \check{e}_y^\star \check{X}_{yx} + \check{e}_z^\star \check{X}_{zx}\right) \qquad (196)$$

$$w_z^{\mathbf{B}} = \check{e}_x^\star e_x\left(\check{X}_{xx}X_{yx} - \check{X}_{xy}X_{xx}\right) + \check{e}_y^\star e_x\left(\check{X}_{yx}X_{yx} - \check{X}_{yy}X_{xx}\right) + \check{e}_z^\star e_x\left(\check{X}_{zx}X_{yx} - \check{X}_{zy}X_{xx}\right) +$$

$$+\check{e}_x^\star e_y\left(\check{X}_{xx}X_{yy}-\check{X}_{xy}X_{xy}\right)+\check{e}_y^\star e_y\left(\check{X}_{yx}X_{yy}-\check{X}_{yy}X_{xy}\right)+\check{e}_z^\star e_y\left(\check{X}_{zx}X_{yy}-\check{X}_{zy}X_{xy}\right)+$$

$$+\check{e}_x^\star e_z\left(\check{X}_{xx}X_{yz}-\check{X}_{xy}X_{xz}\right)+\check{e}_y^\star e_z\left(\check{X}_{yx}X_{yz}-\check{X}_{yy}X_{xz}\right)+\check{e}_z^\star e_z\left(\check{X}_{zx}X_{yz}-\check{X}_{zy}X_{yxz}\right)=$$

$$=e_xX_{yx}\left(\check{e}_x^\star\check{X}_{xx}+\check{e}_y^\star\check{X}_{yx}+\check{e}_z^\star\check{X}_{zx}\right)-e_xX_{xx}\left(\check{e}_x^\star\check{X}_{xy}+\check{e}_y^\star\check{X}_{yy}+\check{e}_z^\star\check{X}_{zy}\right)+$$

$$+e_yX_{yy}\left(\check{e}_x^\star\check{X}_{xx}+\check{e}_y^\star\check{X}_{yx}+\check{e}_z^\star\check{X}_{zx}\right)-e_yX_{xy}\left(\check{e}_x^\star\check{X}_{xy}+\check{e}_y^\star\check{X}_{yy}+\check{e}_z^\star\check{X}_{zy}\right)+$$

$$+e_zX_{yz}\left(\check{e}_x^\star\check{X}_{xx}+\check{e}_y^\star\check{X}_{yx}+\check{e}_z^\star X_{zx}\right)-e_zX_{xz}\left(\check{e}_x^\star\check{X}_{xy}+\check{e}_y^\star\check{X}_{yy}+\check{e}_z^\star\check{X}_{zy}\right)=$$

$$=\left(e_xX_{yx}+e_yX_{yy}+e_zX_{yz}\right)\left(\check{e}_x^\star\check{X}_{xx}+\check{e}_y\check{X}_{yx}+\check{e}_z^\star\check{X}_{zx}\right)-$$

$$-\left(e_xX_{xx}+e_yX_{xy}+e_zX_{xz}\right)\left(\check{e}_x^\star\check{X}_{xy}+\check{e}_y^\star\check{X}_{yy}+\check{e}_z^\star\check{X}_{zy}\right)\tag{197}$$

We have finally obtained:

$$w_x^{\mathbf{B}}=\left(e_xX_{zx}+e_yX_{zy}+e_zX_{zz}\right)\left(\check{e}_x^\star\check{X}_{xy}+\check{e}_y\check{X}_{yy}+\check{e}_z^\star\check{X}_{zy}\right)-$$

$$-\left(e_xX_{yx}+e_yX_{yy}+e_zX_{yz}\right)\left(\check{e}_x^\star\check{X}_{xz}+\check{e}_y^\star\check{X}_{yz}+\check{e}_z^\star\check{X}_{zz}\right)\tag{198}$$

$$w_y^{\mathbf{B}}=\left(e_xX_{xx}+e_yX_{xy}+e_zX_{xz}\right)\left(\check{e}_x^\star\check{X}_{xz}+\check{e}_y\check{X}_{yz}+\check{e}_z^\star\check{X}_{zz}\right)-$$

$$-\left(e_xX_{zx}+e_yX_{zy}+e_zX_{zz}\right)\left(\check{e}_x^\star\check{X}_{xx}+\check{e}_y^\star\check{X}_{yx}+\check{e}_z^\star\check{X}_{zx}\right)\tag{199}$$

$$w_z^{\mathbf{B}}=\left(e_xX_{yx}+e_yX_{yy}+e_zX_{yz}\right)\left(\check{e}_x^\star\check{X}_{xx}+\check{e}_y\check{X}_{yx}+\check{e}_z^\star\check{X}_{zx}\right)-$$

$$-\left(e_xX_{xx}+e_yX_{xy}+e_zX_{xz}\right)\left(\check{e}_x^\star\check{X}_{xy}+\check{e}_y^\star\check{X}_{yy}+\check{e}_z^\star\check{X}_{zy}\right)\tag{200}$$

So using these expression in the (188) we can calculate the mode coupling coefficient for magnetic field fluctuations solely in function of the polarization vector and of the susceptibility tensor.

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